



Taxes and Competition: Evidence from the airline industry

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ABSTRACT

This paper examines whether corporate tax cuts alter product-market competition by differentially affecting firms with high versus low tax burdens. Tax cuts increase after-tax cash flows for profitable firms but provide little immediate benefit to loss-making firms. We study the 1986 Tax Reform Act, which reduced the top corporate tax rate by 12 percentage points, and examine route-level price and quantity data from the U.S. airline industry. We find that in response to the Act, profitable airlines reduce ticket prices by 4.2% relative to their loss-making rivals and gain 3.3 percentage points in market share. These effects are concentrated in routes where loss-making competitors are financially constrained and are accompanied by increased entry by profitable airlines and exit by loss-making airlines. The evidence suggests that taxes can affect competitive outcomes, specifically in our paper by enabling high-tax firms to compete more aggressively with low-tax rivals after tax cuts.

1. Introduction

Corporate income tax policy can have large effects on the competitive landscape in an economy by shaping firms' after-tax cash flows and financial capacity. When tax rates change, firms with greater pre-tax income are often affected to a greater degree because they are more likely to face a higher rate (when rates are graduated), have current-year income subject to tax, and are less likely to have tax loss carryovers. Less profitable firms, by contrast, may experience little change in their financial position. These differences in after-tax cash flows can alter firms' ability to invest, price aggressively, and expand market share, potentially affecting competition in product markets. In this paper, we examine whether corporate tax cuts affect competition among firms that differ in their pre-tax profitability.

Specifically, we examine the differential effect of tax cuts on high-tax firms (proxied by profitable firms) vs. low-tax firms (proxied by loss-making firms) in terms of product market outcomes. Corporate tax cuts increase after-tax cash flows for firms with positive taxable income but not for loss-making firms. Consequently, we hypothesize that profitable firms use these savings to finance price reductions that loss-making rivals cannot readily match, thereby lowering their relative prices and shifting demand—and thus market share—toward profitable firms (holding quality constant). This mechanism is consistent with predation models in which cash-rich firms depress industry profits to strain financially constrained rivals (Fudenberg and Tirole, 1986; Bolton and Scharfstein, 1990). By selectively expanding liquidity only for tax-paying firms, tax cuts widen differences in financial slack and make such strategies more feasible.

Identifying the effect of tax changes on firm decision-making and product market outcomes is challenging because tax policy changes typically occur in response to (or in anticipation of) changing economic conditions, at times making it difficult to distinguish tax effects from business cycle effects. In addition, price and quantity data are rarely available for a broad cross-section of firms around major tax reforms. To address these challenges, we examine how U.S. domestic airlines responded to the Tax Reform Act of 1986 (TRA), which reduced the top corporate income tax rate from 46% to 34%. A key advantage of the airline setting is the availability of

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detailed route-level price and quantity data dating back to the 1980s, when the TRA was enacted. Each airline route constitutes a distinct product market with different sets of competitors (Azar et al., 2018), allowing us to exploit cross-sectional variation in firms' tax positions across markets while filtering out confounding factors such as macroeconomic trends or airline deregulation.¹

We predict that profitable airlines respond to the TRA by reducing (or not increasing as much) ticket prices relative to loss-making airlines, particularly in routes where loss-making airlines are economically important. We also predict that these relative price reductions lead to market share gains for profitable airlines. Our empirical design compares how the same airline adjusts prices across routes that differ in the presence of loss-making competitors. For example, if United Airlines competes with a loss-making rival on one route but with another profitable airline on a different route, we ask whether United Airlines lowers prices more aggressively on the former route following the TRA. We implement this idea using a difference-in-differences design with airline $\times$ route and airline $\times$ year-quarter fixed effects.

We find that the TRA led to a 4.2% decline in average ticket prices of profitable airlines relative to loss-making airlines on routes where loss-making airlines held significant market share. Consistent with this pricing response, profitable airlines gained 3.3 percentage points, on average, in market share in these routes. Economically, for every 10-percentage point increase in the aggregate market share of loss-making airlines, profitable airlines reduce prices by approximately \$2.62 and gain 1.3 percentage points in market share after the TRA. Using the Shevlin (1987, 1990) and Graham (1996) approach to estimate marginal tax rates (MTRs), we calculate that the TRA reduced the average marginal tax rate of profitable airlines by 9.6 percentage points, implying that a one-percentage-point decline in marginal tax rates is associated with a 0.44% price reduction and a 0.34 percentage point increase in market share. Prices and market shares move in parallel prior to the TRA and diverge afterward, mitigating concerns about confounding trends.

These findings align with our central hypothesis that tax cuts confer a competitive advantage to profitable firms by expanding their financial capacity to invest in market share, potentially at the expense of loss-making rivals.² To explore this mechanism further, we examine whether these effects are amplified in routes where loss-making airlines are also financially constrained. Theory suggests that financially constrained firms are more vulnerable to predation because they have fewer resources to sustain losses during a price war (Bolton and Scharfstein, 1990). Consistent with this view, we test whether the relative ticket prices of profitable airlines fall more sharply, and their market shares rise more strongly, after the TRA in routes where the aggregate market share of incumbents that are both loss-making and financially constrained is high.

We follow Zwick and Mahon (2017) and classify an airline as financially constrained in a year if it has not paid a dividend in any of the prior three years and its ratio of cash-to-assets is in the bottom two terciles of the airlines in our sample. We find that the average prices for flights operated by profitable airlines decrease by 4.9% relative to those of loss-making airlines after the TRA in route-year-quarters in which the aggregate market share of financially constrained loss-making incumbent airlines is high. In contrast, we find no significant change in the relative prices for flights operated by profit- and loss-making airlines when the financially constrained loss-making airlines operating in a route have low market share. We also find that profitable airlines gain market share at the expense of loss-making airlines only in route-year-quarters in which the financially constrained loss-making incumbents have relatively high market share. These results are consistent with the idea that the tax savings from the TRA incentivize profitable airlines to cut prices and prey on their financially constrained loss-making rivals that do not benefit from the TRA and do not have the financial resources to survive a prolonged price war.³

We next examine entry and exit patterns following the TRA.⁴ We find that profitable airlines are more likely to enter routes where loss-making airlines are economically important, and such entry effects are almost four times as large in markets where loss-making incumbents are financially constrained. In contrast, loss-making airlines—especially those that are financially constrained—are more likely to exit.

We next examine whether financially constrained profitable airlines are more likely to enter new markets following the TRA. The premise is that the cash tax savings generated by the TRA are particularly valuable for profitable airlines facing binding financing constraints, as these savings relax liquidity constraints and enable expansion into new markets. In contrast, financially unconstrained

¹ The 2017 Tax Cuts and Jobs Act (TCJA) could serve as another shock to corporate tax rates that we could in principle exploit to test our research question. However, none of the publicly listed airlines were in a loss position around the passage of the TCJA, making the setting unviable to test our research question.

² We recognize that tax cuts could lead to differential pricing policies for profit vs. loss firms if the corporate tax incidence falls on consumers (Baker et al., 2024). By focusing on within-airline variation in pricing policies across the routes an airline operates in (with our use of airline $\times$ year-quarter fixed effects and variation in the aggregate market share of loss firms), our research design teases out the effect of the TRA on prices and market share through the competition/predation channel from the tax incidence channel. Our cross-sectional tests and evidence of price reversals after exits (discussed later) also help reinforce our interpretation of the results.

³ We are cognizant that measures of financing constraints available during our sample period suffer from the critique by Farre-Mensa and Ljungqvist (2016), who show that financing constraints proxies used in the literature have limited success partitioning firms into constrained and unconstrained groups. Notwithstanding, we note that the within airline and year-quarter research design we employ mitigates the concern that a correlation between our financing constraints proxy and a firm's growth rate or life cycle (the confounding factors discussed in Farre-Mensa and Ljungqvist (2016)) confounds our results. In addition, in tests tabulated in the online appendix that focus on changes in U.S. state tax rates, we corroborate our findings from the airline setting using the financing constraints proxy proposed by Hoberg and Maksimovic (2015), which is not subject to the critique by Farre-Mensa and Ljungqvist (2016).

⁴ Note that while the analyses of airline ticket prices and market share are conducted at the airline-route-year-quarter level, the analyses of airline entry/exit decisions are conducted at the route-year-quarter level.

airlines would likely have been able to enter new markets even absent the tax cut. Consistent with this prediction, we find that the increase in entry following the TRA is driven by financially constrained profitable airlines, while the entry behavior of financially unconstrained airlines is largely unchanged. This evidence suggests that the TRA relaxes financing constraints for profitable airlines (Edwards et al., 2016; Dobridge, 2021), enabling them to compete more aggressively against loss-making rivals.

Thus far, the evidence presented is not only consistent with profit-making airlines preying on their loss-making rivals post-TRA but also consistent with profit-making airlines using the cash tax savings from the TRA to intensify competition to expand and grow. To distinguish between these interpretations, we draw on theory which suggests that if predation occurs, the predatory firm will raise prices following the exit of its rivals. Consistent with theory, we find that profitable airlines raise prices in route-year-quarters in which a greater number of their loss-making rivals exit the market post-TRA. Such price increases occur only following the exits of loss-making airlines but not following the exits of profitable airlines, suggesting that exits unrelated to past (potentially) predatory actions are not associated with subsequent price increases. Despite the evidence provided, we caveat that we cannot completely disentangle the competition and predation channels, as we are unable to observe the cost structures of airlines.

This paper contributes to the relatively nascent literature on the effect of taxes on competitive outcomes. Recent studies find that tax policy, at times, has different effects on product market competitors, which then leads to differences in operating performance and market share. For example, Donohoe et al. (2022) conclude that the American Jobs Creation Act of 2004 (AJCA) helped U.S. multinational firms (MNCs) gain a competitive edge over domestic rivals by lowering the tax rate on earnings repatriated from foreign subsidiaries. In particular, they find that the liquidity shock from AJCA-induced repatriations enabled MNCs to prey on domestic rivals, as evidenced by a decline in the rivals' earnings. However, a limitation of Donohoe et al. (2022) is that firms' decisions to repatriate earnings in response to the AJCA is endogenous. Similarly, Kim et al. (2021) find that tax cuts in foreign countries hurt U.S. firms with competitors in those countries. Our paper contributes to the literature in two ways. First, our paper shows firms with different pre-tax profits have different product market responses to tax cuts such that tax cuts benefit firms with high pre-tax income at the expense of those with low pre-tax income. Second, the evidence in our paper is based on granular price and market share data from homogeneous product markets. Thus, we contribute to Donohoe et al. (2022) and Kim et al. (2021) by using data on direct competitors and prices across several markets to identify a potential mechanism through which taxes influence product market outcomes.⁵

Our paper also contributes to the literature on the effects of taxes on loss firms (see Hanlon and Heitzman (2010) and Lester and Olbert (2025) for literature reviews). A large number of U.S. corporations report tax losses (Cooper and Knittel, 2010), and prior studies find that loss firms respond differently to changes in tax policy than profitable firms.⁶ For example, Edgerton (2010) and Zwick and Mahon (2017) find that tax policy changes have weaker effects on the investment decisions of loss firms and conclude that firms respond to the tax incentives to invest only when the policy generates immediate cash flows. Wagner et al. (2018) find that Donald Trump's surprise election victory and the associated increase in expectations about a tax cut had a significant effect on the cross-section of stock returns: firms with high effective tax rates and large deferred tax liabilities benefited while those with deferred tax assets resulting from net operating loss (NOL) carryforwards had negative returns.⁷ We extend this literature by examining the differential product-market outcomes for tax-profit vs. tax-loss firms following a tax cut. Understanding economic effects of tax cuts is important when considering tax policy changes. It is tempting to think tax cuts benefit all businesses, incentivize investment for all firms in an economy, etc. However, for some subsets of firms (loss-making in our sample), the tax cuts may have some negative consequences.

Finally, we caveat that our inferences are based on analyzing the airline industry, and as such might not generalize to the broader economy. In our online appendix, we examine changes in state-level corporate income tax rates that help to generalize our results from the airline industry.

2. Background and hypothesis

2.1. The Tax Reform Act of 1986

The Tax Reform Act of 1986 (TRA) passed the House of Representatives in December 1985, the Senate in June 1986, and was signed into law by President Reagan in October 1986. The TRA changed a wide array of aspects of the U.S. Tax Code that can be broadly classified as those affecting 1) individual taxpayers, 2) corporate taxpayers, 3) specific industries or types of income (e.g., taxation of the natural resource, finance, and insurance industries, foreign source income, etc.) and 4) tax shelters (Auerbach and Slemrod, 1997).⁸

⁵ Our paper is also somewhat related Hoopes et al. (2016) and Baugh et al. (2018) who find that online retailers had a competitive advantage at times over brick-and-mortar retailers when online retailers did not have to collect sales tax from their customers while brick-and-mortar retailers were required to do so.

⁶ Edgerton (2010) reports that U.S. loss-corporations reported \$418 billion in losses in their tax returns while profitable corporations reported \$676 billion in profits (in 2002). Cooper and Knittel (2010) use IRS administrative data and report that 45% to 52% of U.S. corporations reported NOLs valued at over \$2.9 trillion from 1993 to 2004. Heitzman and Lester (2022) report that in their sample of (large) firms, 90% have some type of tax loss (federal, state, or foreign).

⁷ While profits create an immediate tax obligation for firms, losses often do not generate immediate tax benefits. In the U.S., the tax law provides that firms with losses can carry them to other years to mitigate the effect of having to report and assess tax in one-year increments. Taxpayers that report a tax loss can carry that loss to offset taxable income in other years; either to prior years and then to future years, or just to future years depending on the laws in place at the time. For our sample period surrounding the TRA, the carryback period for net operating losses was 3 years and the carryforward period was 15 years.

⁸ To our knowledge, the TRA did not include any provisions specific to the airline industry.

For this paper, the relevant changes were those to the corporate tax law. In particular, the TRA lowered the maximum corporate statutory tax rate from 46% to 34%. However, it simultaneously broadened the tax base by eliminating or restricting some deductions.⁹ In addition, the Alternative Minimum Tax was enacted so that companies with ‘excessive’ deductions still have to pay some tax.

Given these offsetting provisions, it is ex ante unclear whether, and to what extent, airlines experienced a decrease in their tax burden post-TRA. Reductions in airlines’ tax rates may have been partially offset by the elimination of deductions and credits (e.g., the repeal of the investment tax credit, modifications to the accelerated depreciation schedule). Our analyses assume that the TRA lowered the decision-relevant tax rate for profitable airlines. To assess this assumption, we examine changes in airlines’ marginal tax rates (MTRs) in our sample following the TRA using the approach of Shevlin (1987, 1990) and Graham (1996). We find that the average profitable (loss-making) airline in our sample experienced a 9.6 (2.8) percentage point decline in its MTR, supporting our use of the TRA as a research setting.¹⁰

In addition, the tax rate reduction lowers the value of unused investment tax credits as well as the future benefit of a firm’s NOL carryforwards. Firms in a tax-loss position are more likely to have unused investment tax credits and NOL carryforwards since they, by definition, did not have positive taxable income to utilize these deductions. Thus, the TRA provides a significant benefit to profitable firms relative to loss firms both directly (by reducing the highest statutory tax rate) and indirectly (by lowering the value of past credits and accumulated deductions). The above said, we note that many airlines lease most of their fleet (Gritta and Lippman, 2010), potentially making the repeal of the investment tax credit less relevant for them.

Putting together our estimate of the drop in the MTR for profit-making airlines and their average pre-tax profit of \$32.6 million, back-of-the-envelope calculations (under several simplifying assumptions) imply that profitable airlines save an additional estimated \$3.1 million in cash annually due to the TRA.¹¹

2.2. Empirical predictions

We argue that tax cuts provide high-tax firms (proxied by firms with pre-tax profits) with a competitive advantage over low-tax firms (proxied by firms with pre-tax losses). Tax cuts generate immediate cash tax savings for profitable firms, expanding their financial slack. These savings can be deployed to lower prices, invest in capacity and market share growth, raise wages or expand employment, and increase payouts. In contrast, firms in a tax loss position receive no immediate cash tax savings and may also experience a decline in the value of their NOL carryforwards, putting them at a relative disadvantage. As a result, in markets where profitable firms compete with loss-making firms, tax cuts create incentives for profitable firms to lower prices and gain market share from their loss-making rivals that do not derive the same benefits from the tax cut.¹²

These incentives are especially strong when loss-making rivals are financially constrained. Bolton and Scharfstein (1990) show that predatory pricing is more likely when competitors face financing frictions. The key intuition is that the cost of aggressive pricing depends on how long prices must remain low: financially unconstrained firms can absorb temporary losses and sustain price wars, whereas financially constrained firms cannot, making them more vulnerable to prolonged competitive pressure.

Finally, we expect profitable firms’ pricing responses to be concentrated in markets where loss-making firms are economically important in the aggregate. Profitable firms have greater scope to expand demand and gain market share when loss-making competitors account for a larger share of the market. Accordingly, the competitive pressure induced by tax cuts is likely to be greatest in markets where loss-making firms have high collective market share.

Tax cuts may also affect competition along the extensive margin. By increasing liquidity for profitable firms, tax cuts can encourage entry into new markets—particularly those in which loss-making incumbents hold substantial aggregate market share. These incentives are strongest when incumbent loss-making firms are financially constrained, as profitable entrants are better positioned to compete aggressively following the tax cut. Increased entry by profitable firms, in turn, is likely to intensify competitive pressure on loss-making incumbents and accelerate their exit, especially when those incumbents lack the financial resources to withstand sustained competition. Finally, consistent with theories of predation, profitable firms may recoup the costs of aggressive pricing by raising prices after their loss-making rivals exit the market.

Our empirical predictions are consistent with analytical evidence in Telser (1966), Fudenberg and Tirole (1986, 1988), and Bolton

⁹ Following the TRA, the highest corporate statutory tax rate under the progressive rate schedule was reduced to 34%. The new tax rates were effective for taxable years beginning on or after July 1, 1987 and a blended rate applied for taxable years that started before July 1, 1987. Thus, for calendar year corporations in the highest tax bracket, the statutory tax rate was 46% before 1987, 40% in 1987 (a 50-50 blended rate of 46% and 34%), and 34% from 1988 forward (until changed again in the early 1990s).

¹⁰ Our estimates of MTRs, which are from Graham (1996), incorporate several features of the tax code (e.g., investment tax credits, NOLs, alternative minimum tax) that may have affected the airline industry.

¹¹ Note that we define loss-making firms as those that have (1) negative pre-tax income and (2) a positive tax loss carryforward (following prior research). Therefore, the definition of profitable firms in the above calculation includes firms with negative pretax income but without a tax loss carryforward. The economic magnitude of the estimate of cash tax savings increases if we limit the sample to firms with positive pretax income (i.e., pre-tax income of \$128.7 million and cash tax saving of \$12.4 million).

¹² The demarcation between profit- and loss-firms is not essential for our inferences. In analyses tabulated in the online appendix, we proxy for high-tax (low-tax) burden firms as those whose pre-tax income is in the top tertile (bottom two terciles), and find that our inferences are unchanged (see Tables OA-5 and OA-6 in the online appendix). Further, we recognize that some airlines might switch between reporting profits and losses from year to year. Thus, we also verify that our results are robust to defining *PROFIT AIRLINE* based on airlines reporting two consecutive years of profits (see Tables OA-7 and OA-8 in the online appendix).

and Scharfstein (1990) who show that cash-rich firms (predators) can force economically efficient but financially constrained rivals (prey) out of business by lowering industry profits and reducing their rivals' cash flows. And prior research finds that firms consider the ex-ante risk of predation when making decisions, and their decisions affect product market outcomes ex post. For example, Haushalter et al. (2007), Fresard (2010), and Hoberg et al. (2014) find that firms facing predatory threats have lower leverage, greater cash holdings, and are more likely to hedge to mitigate underinvestment in the event of predation. Bernard (2016) and Minnis and Shroff (2017) find that financially constrained firms avoid financial disclosure to mitigate predation risk. Other studies focus on the effect of corporate financing decisions on product market outcomes, assuming predation occurs in equilibrium.¹³

Analytical models of predation typically assume that predators temporarily price below marginal cost to force their financially constrained rivals out of business. A key limitation of the empirical literature testing the predation hypothesis (including this study) is that marginal cost data are generally unobservable. As a result, it is difficult to distinguish between intense competition (which results in reduced profits but not losses) and predation (which results in temporary losses for firms as they sell below marginal cost).

2.3. The airline industry

Because we use the airline industry as a laboratory to test our research question, we highlight a few industry features that are particularly relevant. A few years prior to the enactment of the TRA, the Airline Deregulation Act of 1978 fundamentally changed the competitive landscape in the U.S. domestic airline industry by allowing airlines to set prices and enter or exit markets. Despite deregulation, however, two types of barriers to entry continued to moderate competition (see e.g., Borenstein, 1989; Morrison and Winston, 1990, 2000; U.S. Department of Transportation (DOT), 1990, 1999; U.S. General Accounting Office (GAO), 1989, 2001).

One class of entry barriers is marketing-related, such as frequent flyer programs, corporate incentive agreements, and travel agent commission overrides. The second class of barriers is operational. Specifically, access to airport facilities (e.g., boarding gates, ticket counters, baggage handling facilities, and take-off and landing slots), is a necessary input for providing air service and strongly affects airlines' ability to enter new markets. Airlines typically have long-term exclusive-use agreements for boarding gates, which can deter entry by competitors.¹⁴ As a result, airlines can exit markets relatively easily but face substantial obstacles to entry, creating disincentives for airlines to exit routes, especially at slot-constrained airports (Snider and Williams, 2015). Prior studies find that these operating barriers indeed affect competition (DOT, 1999; Ciliberto and Williams, 2010).

The airline setting is well suited to test our predictions for several reasons. First, airline markets are clearly defined by origin-destination pairs, and there are numerous such markets that differ in both the set of competitors and their financial positions. This structure allows us to exploit variation in the financial position of airlines operating in each market and to examine how the same airline responds differently to the TRA across routes. Second, we observe prices and quantities (i.e., the number of passengers) of each airline at the market level, as well as airlines' entry and exit decisions, enabling us to study both intensive- and extensive-margin responses to tax changes. Third, entry into a new market does not require market-specific capabilities or specialized equipment. Rather, airlines rely largely on homogeneous inputs (aircraft, pilots, crew, etc.), allowing them to redeploy their existing resources across markets in response to the TRA, which increases the power of our tests.

Finally, airlines incur substantial fixed costs to serve a market. Anecdotal evidence suggests that fuel costs can account for as much as 50% of an airline's expenses, and these costs are mostly unchanged by the number of passengers on a flight (Mouawad, 2012). As a result, reductions in market share can significantly reduce the operating profitability of a route. To the extent tax cuts enable profitable airlines to reduce airfares relative to loss-making airlines and gain market share, loss-making airlines may face increased pressure to exit such markets and curtail their losses. Consequently, despite the presence of entry barriers discussed above, tax cuts may free up airport facilities both for profitable incumbents to expand their operations in a market, and for new entrants into the market. Taken together, these features make the airline industry highly suitable for testing our predictions.

3. Data

We obtain data on airline ticket prices and market share from the U.S. Department of Transportation's DB1A database, which covers 10% of domestic airline tickets and contains information about the origin, destination, passengers transported, average price, passenger itinerary (e.g., whether the flight is direct or connecting), etc. We use Severin Borenstein's dataset which aggregates information at the airline-route-year-quarter level (Borenstein, 1989).¹⁶ We then combine the datasets for direct and connecting flights (Goolsbee and Syverson, 2008).

We begin our sample selection by identifying all airline-route-year-quarter level observations that have at least 20 passengers for connecting flights and 40 passengers for direct flights each quarter from the first quarter of 1981 to the fourth quarter of 1992, which

¹³ For example, Chevalier (1995a, 1995b), Zingales (1998), and Barrot (2016), among others, find that firms with high leverage, low cash holdings and limited access to short-term liquidity tend to 1) lose market share to their rivals, 2) lower product quality to preserve cash flows, and/or 3) exit the market in situations when predation is likely to have occurred.

¹⁴ On April 5, 2000, President Clinton signed into law AIR-21, which relaxed entry barriers for airlines into new airports. Our sample pre-dates the passage of this law. Presently, only three U.S. airports are slot-constrained.

¹⁵ Airports do not typically charge airlines for gate slots but in gate-constrained airports, airlines with rights to gate slots can sell them to other airlines (and slots in gate-constrained airports were used as collateral during the pandemic).

¹⁶ More information can be obtained here: <https://www.nber.org/research/data/departments-transportation-db1adb1b>.

gives us 1,002,678 initial observations (from 150 unique airlines). We define each route as an origin-destination airport pair without regard to flight direction (Goolsbee and Syverson, 2008; Azar et al., 2018). We drop flights operated by multiple airlines and flights for which we are unable to identify the origin or destination airport. We also aggregate observations at the airline-route-year-quarter level by merging the dataset for connecting and direct flights. These steps reduce our sample to 847,141 observations (and 128 unique airlines). We obtain financial statement information for the airlines in our sample from Compustat. Matching the DB1A data with Compustat results in the loss of 39,671 observations corresponding to 101 unique airlines.¹⁷ A potential concern from dropping unlisted airlines (for whom we cannot identify their profit/loss status) is that the missing market shares of these airlines could be different across routes, and thereby induce a spurious correlation between our variables of interest. We first note that although there are several unlisted airlines in our data, their collective market share is roughly 2%. Notwithstanding their marginal economic footprint, we also verify that our inferences are unchanged when we restrict our sample to routes with just publicly traded airlines (see Table OA-9 in the online appendix).

Our regressions include controls for population and per capita incomes for the endpoints for each route and we obtain the data to compute these variables from the Bureau of Economic Analysis (BEA). We lose 49 observations that are missing BEA data, leaving us with 807,421 observations. Since our focus is on explaining differences in the prices charged by profitable and loss-making airlines operating in the same route, and such variation is absent in single-airline routes, we drop route-year-quarters that are served by only one airline. This restriction leaves us with 580,588 observations. In addition, dropping route-year-quarters that do not have at least one profitable airline and at least one loss-making airline leaves us with 375,298 observations.¹⁸ Finally, restricting our sample to the period from 1982 Q1 to 1992 Q4 leaves us with a final sample of 364,812 observations (27 unique airlines). Table 1 presents the sample selection procedure.

4. Research design and descriptive statistics

We estimate the following difference-in-differences regression to test our predictions:

$$y_{i,r,t} = \beta_1 \text{MKT IMPORTANCE OF LOSS AIRLINES}_{r,t-1} + \beta_2 \text{MKT IMPORTANCE OF LOSS AIRLINES}_{r,t-1} \times \text{PROFIT AIRLINE}_{i,t-1} + \beta_3 \text{MKT IMPORTANCE OF LOSS AIRLINES}_{r,t-1} \times \text{POST86}_t + \beta_4 \text{MKT IMPORTANCE OF LOSS AIRLINES}_{r,t-1} \times \text{PROFIT AIRLINE}_{i,t-1} \times \text{POST86}_t + \alpha_i \times \alpha_r + \alpha_i \times \alpha_t + \alpha_t \times \text{LN(DISTANCE)} + \gamma'X + \varepsilon_{i,r,t} \quad (1)$$

Where i , r , and t index airlines, routes, and year-quarters, respectively; $y_{i,r,t}$ is either the natural logarithm of average fare charged by an airline in a route-year-quarter (LN(PRICE)) or airline market share in a route-year-quarter, where market share is measured as the number of passengers flying by airline 'i' in route 'r' and period 't' scaled by the total number of passengers flying that route in the same period (MARKET SHARE). We use weighted least squares when the dependent variable is LN(PRICE) , with weights based on the average number of passengers traveling with an airline in a route over the sample period (Goolsbee and Syverson, 2008; Azar et al., 2018).

Since profits create an immediate tax obligation for firms while losses do not, we proxy for high-tax burden airlines as those that report a pre-tax profit in a year, while low-tax burden airlines are those with a tax loss. Specifically, we construct an indicator variable, *PROFIT AIRLINE*, that equals one for airlines that did not incur a tax loss in a year. Following Mills et al. (2003), we classify an airline-year as being in a tax loss position if it reports negative pre-tax income in a fiscal year and also has an NOL carryforward. *MKT IMPORTANCE OF LOSS AIRLINES* measures the economic importance of loss-airlines (in aggregate) in a route and year-quarter, which we proxy for using the aggregate market share of all tax-loss airlines operating in the route-year-quarter. *POST86* is an indicator variable that equals one for calendar years 1986 and onwards. The TRA passed the House of Representatives in December 1985, the Senate in June 1986, and was signed into law by President Reagan in October 1986.¹⁹ The new statutory tax rate took effect for taxable years beginning on or after July 1, 1987.

Our interest lies in the period when uncertainty about the enactment of the TRA and the reduction in the statutory tax rate had passed, which we believe to have occurred after the TRA passed the House. We expect that any treatment effect of the TRA on the behavior of profit- and loss-making airlines would occur soon after expectations were set that the bill would become law rather than following its effective date. Prior research finds that firms respond to known changes in tax rates prior to their effective dates to take maximum advantage of the tax rules (Scholes et al., 1992). When it became known that the TRA would become law and lower tax rates, it was optimal for firms to shift expenses to the pre-TRA period and shift income to the post-TRA period. Investments aimed at expanding market share (lowering price, entering new routes, increasing advertising, etc.) reduce contemporaneous taxable income

¹⁷ By matching DB1A data to Compustat, our sample is restricted to airlines that are publicly traded. While this sample restriction leads to a large drop in the number of unique airlines in our sample, the drop in the number of airline-route-year-quarter observations is relatively small. The finding that publicly traded airlines dominate most markets is not surprising, considering the significant capital investment required to operate an airline.

¹⁸ Our sample restriction related to single-airline routes is consistent with that in Borenstein (1989), who examines whether airline dominance in a route leads to differences in ticket prices charged by the dominant airline vs. other airlines operating in the same route. We relax this assumption when we examine airline entry and exit patterns. In addition, we verify that our results are robust to retaining single-airline routes in our sample (see Table OA-10 in the online appendix).

¹⁹ Most of the provisions of the TRA were effective on January 1, 1987; a few were retroactive to January 1, 1986, and some were phased in over the next few years (Wakefield, 1987).

Table 1
Sample selection.

	Sample composition	Obs. Dropped	Obs. Remaining	Airline-year-quarters remaining	Airlines remaining
(1)	Airline-route-year-quarter observations with more than 20 (40) passengers for connected (direct) flights in a quarter		1,002,678	2,311	150
(2)	Less: Flights operated by multiple airlines	43,552	959,126	2,078	130
(3)	Less: Duplicate observations identified by merging connecting flights with direct flights for each airline-route-year-quarter	110,456	848,670	2,070	129
(4)	Less: Observations with unidentified airports	1,529	847,141	2,064	128
(5)	Less: Observations without a Compustat match in period t and $t-1$	39,671	807,470	892	27
(6)	Less: Observations with missing population and income data	49	807,421	892	27
(7)	Less: Route-year-quarter with fewer than two airlines	226,833	580,588	890	27
(8)	Less: Route-year-quarter without at least one profitable and one loss-making airline	205,290	375,298	845	27
(9)	Restricting the sample period to 1982 Q1 - 1992 Q4.	10,486	364,812	777	27
	Final sample of airline-route-year-quarter observations available for analyses		364,812	777	27

Notes: This table presents the sample selection procedure of our main analyses employing data from the airline industry and the Tax Reform Act of 1986.

and therefore are particularly attractive to undertake before the tax cut, while their competitive benefits materialize afterward. Consistent with this logic, we expect profitable airlines to have begun lowering prices shortly after uncertainty about the TRA's passage was resolved, rather than waiting until the new statutory rate became effective. This timing aligns with our empirical design, which treats the post-1985 period as the relevant window for identifying the competitive effects of the TRA.

Equation (1) includes a rich set of fixed effects and control variables following Azar et al. (2018) to isolate within-airline, within-route variation in pricing and market share around the TRA. Specifically, we include airline $\times$ year-quarter fixed effects, which absorb all time-varying and time-invariant factors affecting an airline's behavior across all routes it operates, including firm-wide cost shocks, changes in aggregate demand for air travel, and industry-wide events such as the Airline Deregulation Act of 1978. This structure allows for heterogeneous responses across airlines to macroeconomic and regulatory shocks, including any aggregate demand effects associated with the TRA.

We also include airline $\times$ route fixed effects, which control for all time-invariant determinants of an airline's pricing policies in a given route, such as persistent differences in market power arising from hub locations, frequent flyer programs, network structure, and route-specific cost advantages. These fixed effects ensure that identification comes from changes within the same airline-route pair over time. To account for time-varying route-level cost shocks, we include year-quarter $\times$ log(distance) fixed effects, which flexibly control for differential exposure to fuel price fluctuations across short- and long-haul routes. In addition, we control for time-varying market conditions, including the number of airlines operating in a route, the share of passengers traveling on connecting flights at both the airline-route and route levels, and local demand conditions measured by the log of average population and per-capita income at the route endpoints. We double-cluster standard errors at the airline and year-quarter levels; our inferences are robust to clustering at the route and year-quarter levels (see Table OA-12).

A key feature of our research design is that it exploits *cross-sectional variation in the economic importance of loss-making airlines at the route-year-quarter level*, which is uniquely tied to our theory. Profitable airlines have greater incentives to lower prices and expand market share in routes where loss-making competitors collectively hold larger market shares. By comparing how the same airline adjusts pricing and market share across routes that differ in this characteristic, our design further mitigates concerns about omitted variables and endogeneity.

One potential alternative explanation is tax incidence: if corporate tax cuts are partially passed through to consumers, profitable airlines may lower prices after the TRA simply because their costs fall, regardless of competitive conditions. However, such a mechanism would predict price reductions by profitable airlines across all routes they operate. In contrast, our identification relies on within-airline differences across routes that vary in the presence and importance of loss-making competitors. The inclusion of airline $\times$ year-quarter fixed effects ensures that any uniform pass-through of tax savings at the airline level is absorbed, allowing us to separately identify competitive responses linked to rival characteristics.²⁰ A final concern is that *MARKET SHARE* may be mechanically related to *MKT IMPORTANCE OF LOSS AIRLINES*, although the latter is lagged. To the extent any mechanical relation is stable over time, our difference-in-differences design differences it out. Moreover, such a mechanical relation is unlikely to vary discretely between the pre- and post-TRA periods or to differ systematically with the financial constraints of loss-making airlines, as our hypotheses predict.

Taken together, the coefficient on *MKT IMPORTANCE OF LOSS AIRLINES* $\times$ *PROFIT AIRLINE* $\times$ *POST86* captures within-airline changes in pricing and market share across routes for profitable airlines relative to loss-making airlines after the TRA, interacted with cross-sectional variation in the lagged economic importance of loss-making competitors. Identification comes from comparing how the same profitable airline adjusts behavior across routes with different competitive environments before vs. after the TRA,

²⁰ Our research setting precludes us from testing the tax incidence channel because we lack variation on when consumers would be more or less likely to bear the tax burden.

holding constant airline-level and route-level confounds.

Table 2, Panel A reports descriptive statistics for our sample at the airline-route-year-quarter level. The average fare per passenger across all routes is \$164 (natural log transformed in the table), which amounts to \$355 in 2020 dollars. The average airline has 27% market share in each route that it operates in while the aggregate market share of loss airlines in the average route is 31%. Approximately one-third of route-year-quarters are operated by loss-airlines. The average number of passengers flying per airline-route-year-quarter is 3,186 and the average number of airlines operating in a route-year-quarter is 4.6. On average, around four-fifths of an airline's passengers use connecting flights and around two-thirds of passengers in a given route use connecting flights. The average population is 3.1 million and the average income is approximately \$18,500 (which is \$39,600 in 2020 dollars) in the metro areas at the route endpoints. These statistics are consistent with Azar et al. (2018).

Panels B and C in Table 2 present the descriptive statistics above (which are at the airline-route-year-quarter level) separately for profitable and loss-making airlines. Since our identification exploits variation in airlines' profit vs. loss status, we make additional comparisons at the airline-level that are untabulated. Airlines that report a profit throughout our sample period are larger in size than those that report a loss in at least 50% of the years over our sample period – profitable airlines report average total assets of \$2.9 billion and on average sold 10.6 million tickets each year compared to \$1.6 billion and 6.3 million for those with losses. Loss airlines have higher growth rates than profitable airlines (41% vs. 21% average annual asset growth and 27% vs. 18% average annual passenger growth over our sample period), and went public two years later, on average. These differences between profitable and loss-making airlines suggest that there might be some lifecycle effect on ticket prices and market share, and as such reinforce the importance of controlling for airline $\times$ year-quarter fixed effects. In our sample, only five airlines are low-cost carriers (per the definition of the International Civil Aviation Organization), and of these, two airlines reported tax-losses in more than 50% of our sample period.

We recognize that variation in *MKT IMPORTANCE OF LOSS AIRLINES* affects the interpretation of our results. To provide insight into the nature of such variation, we sort routes into deciles based on *MKT IMPORTANCE OF LOSS AIRLINES* and present several descriptive characteristics of the routes in each decile. Table 3 shows that loss-making airlines operate across a wide variety of route types – those with high and low passenger volume, and long and short distances. That said, routes in the bottom three deciles of *MKT IMPORTANCE OF LOSS AIRLINES* have relatively greater passenger volume, and those in the top three deciles have shorter average distances, cheaper average ticket prices, and fewer competing airlines. While it is unclear whether such heterogeneity biases the interpretation of our results, in analyses tabulated in the online appendix (Table OA-13), we verify that our results are robust to matching routes with high *MKT IMPORTANCE OF LOSS AIRLINES* to those with low *MKT IMPORTANCE OF LOSS AIRLINES*, based on several route characteristics using entropy balancing.

5. Empirical results

This section presents our empirical findings. Section 5.1 examines how the TRA affects ticket prices and market shares of profitable relative to loss-making airlines, and explores heterogeneity in these effects based on the financial constraints of loss-making competitors. Section 5.2 analyzes entry and exit responses to the TRA, focusing on whether profitable airlines expand into, and loss-making airlines exit from, markets where the competitive effects are strongest. Finally, Section 5.3 studies post-exit pricing behavior to assess whether profitable airlines raise prices following the exit of loss-making rivals, as predicted by theories of predation.

5.1. Effect of TRA on prices and market share

Before presenting our regression analyses, we provide univariate descriptive evidence on the differences in ticket prices charged by profitable vs. loss-making airlines in routes with high and low market importance of loss airlines, before and after the TRA. Specifically, we compute the average ticket price difference for profitable and loss-making airlines in each route-year-quarter. We then purge variation in the price difference attributable to factors such as frequent flyer programs, customer service quality, flight frequency, reward systems for travel agents that pay bonuses for booking more travel with a particular airline to the exclusion of others, etc. (see e. g., Borenstein, 1989). In particular, we regress the price difference on route, year-quarter, and year-quarter $\times$ distance fixed effects, as well as the average population, income, and percent of connecting passengers flying the route. The residuals from this regression represent the portion of price difference unexplained by route and time-specific characteristics that determine the base price gap between profitable and loss-making airlines in the same route-year-quarter.

Analyzing these residuals, we find that, before the TRA, the ticket prices of profitable firms are, on average, \$0.58 lower than those of their loss-making rivals in routes where loss-making airlines have a high market share. In the post-TRA period, the average ticket price of profitable airlines becomes \$3.38 lower than that of their loss-making rivals. These dollar differences correspond to 0.4% and 2.1% of the average ticket price (\$163.8) during our sample period, respectively. By contrast, in routes where loss-making airlines have a low market share, we do not observe a significant difference in the average ticket prices of profitable and loss-making airlines, which is \$0.06 and \$0.38 in the pre- and post-TRA period, respectively. These results provide descriptive evidence that, post-TRA, profitable airlines lower prices relative to loss-making airlines in routes where the loss-making airlines have high market share, consistent with our proposed predation mechanism.

We next turn to our regression evidence. Table 4, Panel A reports estimates of equation (1). Our coefficient of interest is the triple interaction *MKT IMPORTANCE OF LOSS AIRLINES* $\times$ *PROFIT AIRLINE* $\times$ *POST86*, which captures whether profitable airlines adjust prices and market share differentially after the TRA in routes where loss-making competitors are economically important. When the dependent variable is *LN(PRICE)*, this coefficient measures the post-TRA change in ticket prices for profitable airlines relative to loss-making airlines operating in the same route and period, interacted with the aggregate market share of loss-making airlines in that

Table 2
Descriptive statistics.

Panel A: Full sample						
Variables	Mean	SD	P25	P50	P75	N
<i>LN(PRICE)</i>	5.045	0.341	4.860	5.071	5.268	364,812
<i>MARKET SHARE</i>	0.267	0.231	0.073	0.200	0.411	364,812
<i>PROFIT AIRLINE</i>	0.671	0.470	0.000	1.000	1.000	364,812
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> ⁺	0.309	0.258	0.103	0.241	0.453	364,812
<i>AVG. PASSENGERS</i>	3,186	6,533	525	976	2,309	364,812
<i>#AIRLINES</i>	4.598	2.042	3.000	4.000	6.000	364,812
<i>%CONNECTING - AIRLINE</i>	0.809	0.363	1.000	1.000	1.000	364,812
<i>%CONNECTING - ROUTE</i>	0.652	0.378	0.263	0.837	1.000	364,812
<i>LN(POPULATION)</i>	14.662	0.760	14.126	14.627	15.138	364,812
<i>LN(INCOME)</i>	9.810	0.187	9.676	9.833	9.954	364,812
Panel B: Subsample of profitable airlines						
Variables	Mean	SD	P25	P50	P75	N
<i>LN(PRICE)</i>	5.061	0.342	4.873	5.088	5.288	244,622
<i>MARKET SHARE</i>	0.261	0.229	0.069	0.193	0.403	244,622
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> ⁺	0.286	0.247	0.093	0.218	0.416	244,622
<i>AVG. PASSENGERS</i>	3,362	6,773	553	1,040	2,437	244,622
<i>#AIRLINES</i>	4.804	2.021	3.000	4.000	6.000	244,622
<i>%CONNECTING - AIRLINE</i>	0.811	0.359	1.000	1.000	1.000	244,622
<i>%CONNECTING - ROUTE</i>	0.646	0.375	0.262	0.813	1.000	244,622
<i>LN(POPULATION)</i>	14.691	0.749	14.159	14.647	15.164	244,622
<i>LN(INCOME)</i>	9.814	0.182	9.686	9.835	9.955	244,622
Panel C: Subsample of loss-making airlines						
Variables	Mean	SD	P25	P50	P75	N
<i>LN(PRICE)</i>	5.011	0.337	4.835	5.039	5.227	120,190
<i>MARKET SHARE</i>	0.279	0.234	0.083	0.215	0.427	120,190
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> ⁺	0.354	0.273	0.132	0.295	0.523	120,190
<i>AVG. PASSENGERS</i>	2,828	6,001	474	855	2,044	120,190
<i>#AIRLINES</i>	4.180	2.021	3.000	4.000	5.000	120,190
<i>%CONNECTING - AIRLINE</i>	0.805	0.370	1.000	1.000	1.000	120,190
<i>%CONNECTING - ROUTE</i>	0.665	0.383	0.264	0.888	1.000	120,190
<i>LN(POPULATION)</i>	14.605	0.780	14.050	14.578	15.051	120,190
<i>LN(INCOME)</i>	9.800	0.195	9.653	9.829	9.953	120,190

Notes: This table presents the descriptive statistics for the variables used in our analyses conducted using the airline setting and the 1986 Tax Reform Act. Panel A reports the descriptive statistics for our sample at the airline-route-year-quarter level. Panels B and C present the descriptive statistics separately for profitable and loss-making airlines. In the tables above, + indicates that the variable is standardized to have a mean of zero and a standard deviation of one in our regression analyses. See [Appendix A](#) for variable definitions.

Table 3
Route characteristics across deciles of loss-making airlines' market importance.

<i>MKT IMPORTANCE OF LOSS AIRLINES</i>		Number of passengers	Distance	Price (in \$)	#AIRLINES	%CONNECTING - ROUTE
Decile rank	Mean					
1	0.01	20,886	1,097	166	3.93	0.51
2	0.08	17,716	1,277	175	4.78	0.54
3	0.13	13,369	1,326	173	4.46	0.70
4	0.20	11,811	1,306	172	4.16	0.76
5	0.26	11,135	1,232	169	3.88	0.78
6	0.33	11,248	1,152	165	3.69	0.78
7	0.41	10,174	1,082	162	3.42	0.78
8	0.52	10,073	1,024	159	3.24	0.77
9	0.66	10,765	984	158	3.15	0.70
10	0.91	12,515	907	155	2.75	0.62

Notes: This table presents descriptive statistics for route characteristics across deciles of *MKT IMPORTANCE OF LOSS AIRLINES*, including the number of passengers, distance, average ticket price, number of airlines operating in the route, and percentage of passengers traveling on connecting flights. See [Appendix A](#) for variable definitions.

route. Intuitively, it tests whether profitable airlines cut prices more aggressively after the tax cut precisely in markets where they stand to gain the most by displacing loss-making rivals.

Columns 1 and 2 present results for ticket prices. Column 1 reports estimates without time-varying controls, while column 2 includes route-level and airline-route-level controls. In both specifications, the coefficient for *MKT IMPORTANCE OF LOSS*

Table 4

Effect of tax rate reductions on the ticket prices and market shares of loss-making vs. profitable airlines.

Panel A: Static Analyses				
Dependent Variable:	LN(PRICE)		MARKET SHARE	
	(1)	(2)	(3)	(4)
	Coef.	Coef.	Coef.	Coef.
	t-Stat.	t-Stat.	t-Stat.	t-Stat.
MKT IMPORTANCE OF LOSS AIRLINES	−0.009 (−0.954)	−0.010 (−1.016)	0.067*** (5.162)	0.059*** (5.294)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE	0.013 (1.407)	0.010 (1.186)	−0.096*** (−6.835)	−0.086*** (−7.043)
MKT IMPORTANCE OF LOSS AIRLINES × POST86	0.038*** (3.809)	0.037*** (4.048)	−0.027** (−2.144)	−0.024** (−2.100)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE × POST86	−0.046*** (−3.442)	−0.042*** (−3.559)	0.037** (2.736)	0.033** (2.719)
#AIRLINES		−0.011*** (−3.075)		−0.029*** (−22.916)
%CONNECTING - AIRLINE		−0.007 (−0.350)		−0.277*** (−19.555)
%CONNECTING - ROUTE		0.125*** (6.904)		0.185*** (9.564)
LN(POPULATION)		−0.359** (−2.155)		−0.006 (−0.083)
LN(INCOME)		0.040 (0.162)		0.003 (0.075)
<i>Fixed effects</i>				
Airline × Route	Included	Included	Included	Included
Airline × Year-Quarter	Included	Included	Included	Included
Year-Quarter × LN(DISTANCE)	Included	Included	Included	Included
R-Squared	89.9%	90.1%	82.2%	85.7%
No. of Observations	364,812	364,812	364,812	364,812
Panel B: Dynamic Analyses				
Dependent Variable:	LN(PRICE)		MARKET SHARE	
	(1)	(2)	(3)	(4)
MKT IMPORTANCE OF LOSS AIRLINES	0.000 (0.044)	−0.002 (−0.204)	0.069*** (9.627)	0.057*** (8.775)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE	0.004 (0.386)	0.002 (0.139)	−0.087*** (−11.416)	−0.077*** (−11.157)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR82	0.002 (0.186)	0.005 (0.557)	0.011 (0.833)	0.014 (1.360)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR83	−0.004 (−0.284)	−0.002 (−0.161)	−0.005 (−1.032)	0.002 (0.542)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR84	−0.033* (−2.035)	−0.031* (−1.998)	−0.009 (−0.742)	−0.004 (−0.335)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR86	−0.015 (−1.574)	−0.014 (−1.514)	−0.028* (−1.768)	−0.023 (−1.499)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR87	0.022** (2.338)	0.025** (2.708)	−0.020*** (−3.206)	−0.015*** (−3.397)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR88	0.030*** (3.420)	0.031*** (3.746)	−0.020** (−2.542)	−0.015* (−1.868)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR89	0.031*** (3.981)	0.033*** (4.556)	−0.033** (−2.190)	−0.025* (−1.826)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR90	0.019* (2.001)	0.020* (1.968)	−0.028** (−2.457)	−0.023** (−2.253)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR91	0.044*** (6.255)	0.046*** (6.914)	−0.034*** (−3.285)	−0.025** (−2.739)
MKT IMPORTANCE OF LOSS AIRLINES × YEAR92	0.032** (2.348)	0.034** (2.681)	−0.035** (−2.655)	−0.026** (−2.289)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE × YEAR82	0.004 (0.284)	0.004 (0.300)	−0.036** (−2.378)	−0.029** (−2.359)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE × YEAR83	0.005 (0.295)	0.007 (0.360)	−0.007 (−1.047)	−0.008 (−1.311)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE × YEAR84	0.024 (1.138)	0.023 (1.185)	−0.005 (−0.425)	−0.003 (−0.285)
MKT IMPORTANCE OF LOSS AIRLINES × PROFIT AIRLINE × YEAR86	−0.023 (−0.972)	−0.017 (−0.748)	0.033** (2.292)	0.029* (2.018)

(continued on next page)

Table 4 (continued)

Panel B: Dynamic Analyses				
Dependent Variable:	LN(PRICE)		MARKET SHARE	
	(1)	(2)	(3)	(4)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>YEAR87</i>	−0.032** (−2.092)	−0.029* (−1.848)	0.024*** (2.987)	0.023*** (3.964)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>YEAR88</i>	−0.042*** (−3.046)	−0.038** (−2.743)	0.016* (1.715)	0.015 (1.594)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>YEAR89</i>	−0.024 (−1.630)	−0.020 (−1.545)	0.027* (1.754)	0.025* (1.802)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>YEAR90</i>	−0.025 (−1.306)	−0.020 (−1.215)	0.019 (1.598)	0.020* (1.796)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>YEAR91</i>	−0.044*** (−3.682)	−0.041*** (−3.680)	0.027** (2.711)	0.025*** (2.926)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>YEAR92</i>	−0.044** (−2.162)	−0.041** (−2.206)	0.038*** (2.986)	0.034*** (3.259)
Airline-route level time varying controls	Not included	Included	Not included	Included
Fixed effects: Airline × Route, Airline × Year-Quarter, and Year-Quarter × LN(DISTANCE)	Included	Included	Included	Included
R-Squared	90.0%	90.1%	82.2%	85.7%
No. of Observations	364,812	364,812	364,812	364,812

Notes: This table presents the results from regressions of 1) the natural logarithm of average ticket prices charged by airlines in each route-year-quarter or 2) airlines' market shares in each route-year-quarter on interaction terms between the aggregate market share of loss-making airlines operating in the route-year-quarter, an indicator variable that equals one for whether the airline made a tax profit the previous year (*PROFIT AIRLINE*), and an indicator variable that equals one for the post-1986 TRA period (*POST86*), and control variables. Note that the main effects of *POST86*, *PROFIT AIRLINE*, and *POST86* × *PROFIT AIRLINE* are subsumed by our fixed effects. We use weighted least squares when the dependent variable is ticket prices, with weights based on the average number of passengers traveling with an airline in a route over the entire sample period. The regressions in Panel A estimate a single post-treatment indicator for the period after the anticipated enactment of the TRA 1986, and the regressions in Panel B estimate separate coefficients for each of the years in our sample, excluding 1985 which serves as the base year. See Appendix A for variable definitions. Standard errors are clustered at the airline and year-quarter levels. ***, **, and * denote statistical significance at the two-tailed 1, 5, and 10 percent levels, respectively.

AIRLINES × *PROFIT AIRLINE* × *POST86* is negative and statistically significant (−0.046 in column 1 and −0.042 in column 2). These estimates imply that, following the TRA, profitable airlines reduced ticket prices by approximately 4.2–4.6 percent relative to loss-making airlines on the same routes in markets where loss-making airlines are economically important. To better understand the magnitude of these effects, consider the following. A 10-percentage point increase in the aggregate market share of loss-making airlines in a route is associated with an additional 1.6 percent post-TRA price reduction by profitable airlines, relative to loss-making competitors. At the sample mean ticket price, this corresponds to a reduction of approximately \$2.62 per ticket.

Columns 3 and 4 report the results when the dependent variable is *MARKET SHARE*. In column 3, which excludes time-varying controls, the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* × *PROFIT AIRLINE* × *POST86* is 0.037 (t-stat. = 2.74). In column 4, which includes route-level and airline-route-level controls, the coefficient is slightly smaller but remains statistically significant at 0.033 (t-stat. = 2.72). These estimates indicate that, following the TRA, profitable airlines gain economically meaningful market share from their loss-making rivals in routes where loss-making airlines are important competitors. Specifically, profitable airlines increase their market share by approximately 3.3–3.7 percentage points relative to loss-making airlines operating in the same route and period. In economic terms, a 10-percentage point increase in the aggregate market share of loss-making airlines in a route is associated with a 1.3 percentage point increase in a profitable airline's post-TRA market share, relative to loss-making airlines and relative to the pre-TRA period. This pattern is consistent with profitable airlines using the cash-flow advantages created by the TRA to expand demand and capture market share from competitors that do not benefit from the tax cut.

These results are robust to including additional fixed effects at the route × year-quarter level to control for route-level supply and demand shocks, following Bloomfield (2024) (see Table OA-11 in the online appendix).²¹ We also verify that our results are robust to using alternative proxies for high vs. low tax airlines based on their pre-tax profits (see Tables OA-5 and OA-6) and to clustering standard errors by route and year-quarter (see Table OA-12).

We obtain data on firms' marginal tax rates (MTRs) – computed as the ratio of the estimate of the additional taxes payable on an incremental dollar of earnings to the dollar of earnings – from John Graham's website.²² Using these estimates, we compute the change in the MTR for profitable airlines. We find that the MTR for profitable airlines declined by 9.6 percentage points. Thus, the results in

²¹ We exclude route × year-quarter fixed effects from our main specification to 1) stay consistent with the research design in Azar et al. (2018) and 2) allow the entry of new airlines and exit of incumbents to affect prices and market share as it is part of the mechanism through which we expect the TRA to affect these variables.

²² <https://people.duke.edu/~jgraham/taxform.html>.

Table 4, Panel A suggest that a one-percentage point decline in the MTR for profitable airlines leads to a 0.44% decrease in the ticket prices of profit-making airlines relative to those of loss-making airlines, and a 0.34 pp increase in the market share of profit-making airlines.²³ The coefficients for the control variables have the expected signs and are consistent with prior research: for example, a larger number of competitors (*#AIRLINES*) is associated with lower prices and market shares for individual airlines and the share of passengers traveling with a connecting flight in the market (*%CONNECTING – ROUTE*) is positively associated with prices and market share.

In Table 4, Panel B, we replace the single post-TRA indicator (*POST86*) with a full set of calendar-year indicators, omitting 1985 as the base year. This specification allows us to trace the dynamic evolution of prices and market shares around the TRA and to assess whether profitable and loss-making airlines exhibit parallel trends prior to the reform. Columns 1 and 2 show that the average ticket prices of profitable airlines begin to decline relative to those of loss-making airlines in 1986, with the difference becoming economically and statistically meaningful thereafter. In particular, relative prices remain 2.0 to 4.1 percent lower for profitable airlines throughout the post-TRA period (column 2). Importantly, we do not observe systematic relative price differences in the years immediately preceding the TRA. Columns 3 and 4 present analogous results for market share. Profitable airlines begin to gain market share from loss-making rivals starting in 1986, with no evidence of differential trends in the pre-TRA period. Fig. 1 graphically illustrates these dynamics using the estimates from columns 2 and 4. Taken together, the dynamic patterns in prices and market shares support our identifying assumption of parallel pre-trends and strengthen the interpretation that the observed post-1986 divergence reflects the competitive effects of the TRA rather than coincident macroeconomic conditions or lingering effects of airline deregulation.

We next examine whether the pricing response of profitable airlines is stronger in markets where their loss-making rivals are financially constrained. Theoretical models of predation predict that financially constrained firms are particularly vulnerable to aggressive pricing by cash-rich competitors because they lack the liquidity to sustain losses during a price war (Bolton and Scharfstein, 1990). When rivals cut prices, financially constrained firms face a stark tradeoff: match the price cut and further strain already limited cash flows, or maintain prices and risk losing market share. Tax cuts intensify this asymmetry by increasing liquidity only for profitable firms. As a result, we predict that the TRA incentivizes profitable airlines to lower ticket prices primarily in routes where loss-making competitors are both economically important and financially constrained. To test this prediction, we partition the sample into route-year-quarters in which the aggregate market share of financially constrained loss-making airlines is in the top three quartiles versus the bottom quartile, and re-estimate equation (1) within each subsample.

Following Zwick and Mahon (2017), we classify an airline as financially constrained in a given year-quarter if it has not paid dividends in the prior three years and its cash-to-assets ratio is in the bottom two terciles of the airline distribution. While this proxy is necessarily imperfect, newer text-based measures of financing constraints are unavailable during our sample period. Nevertheless, our identification strategy mitigates concerns that unobserved firm characteristics drive the results. In particular, the inclusion of airline $\times$ year-quarter fixed effects absorb time-varying airline-level factors common across routes, isolating within-airline differences in competitive responses across markets.

Table 5 presents the results. When the dependent variable is *LN(PRICE)*, the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES $\times$ PROFIT AIRLINE $\times$ POST86* is -0.049 ($t\text{-stat.} = -3.39$) in the regression estimated on the subsample of route-year-quarters where financially constrained loss-making airlines have high market share (column 1). In contrast, the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES $\times$ PROFIT AIRLINE $\times$ POST86* is -0.019 ($t\text{-stat.} = -1.25$) in the regression estimated on the subsample of route-year-quarters where financially constrained loss-making airlines have low market share (column 2). The difference in coefficients across the two subsamples is statistically significant ($p\text{-value} = 0.061$).

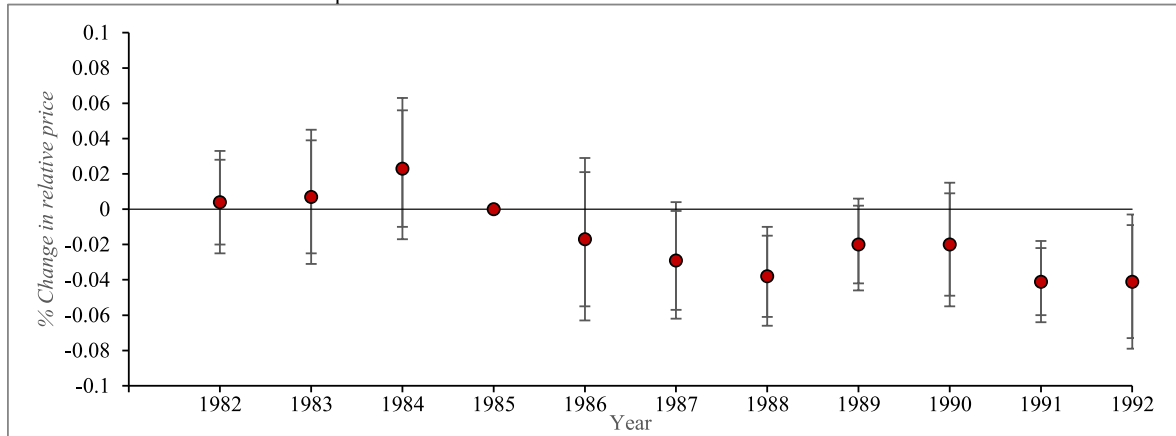
Columns 3 and 4 present analogous results for *MARKET SHARE*. Profitable airlines gain market share from loss-making rivals only in routes where financially constrained loss-making airlines are economically important. Specifically, the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES $\times$ PROFIT AIRLINE $\times$ POST86* is 0.038 ($t\text{-stat.} = 2.71$) in route-year-quarters where financially constrained loss-making airlines have high market share (column 3) and is 0.014 ($t\text{-stat.} = 1.24$) in route-year-quarters where financially constrained loss-making airlines have low market share (column 4).²⁴

Taken together, these results indicate that the competitive effects of the TRA are concentrated in markets where loss-making rivals are both economically important and financially constrained. Economically, a 10-percentage point increase in the aggregate market share of financially constrained loss-making airlines is associated with an additional 1.9 percent post-TRA price reduction by profitable airlines (which corresponds to approximately \$3.11) and a 1.47 percentage point increase in their market share. These effects are approximately 15 percent larger than the average effects documented in Table 4, consistent with financially constrained rivals being less able to withstand aggressive pricing following the tax cut.

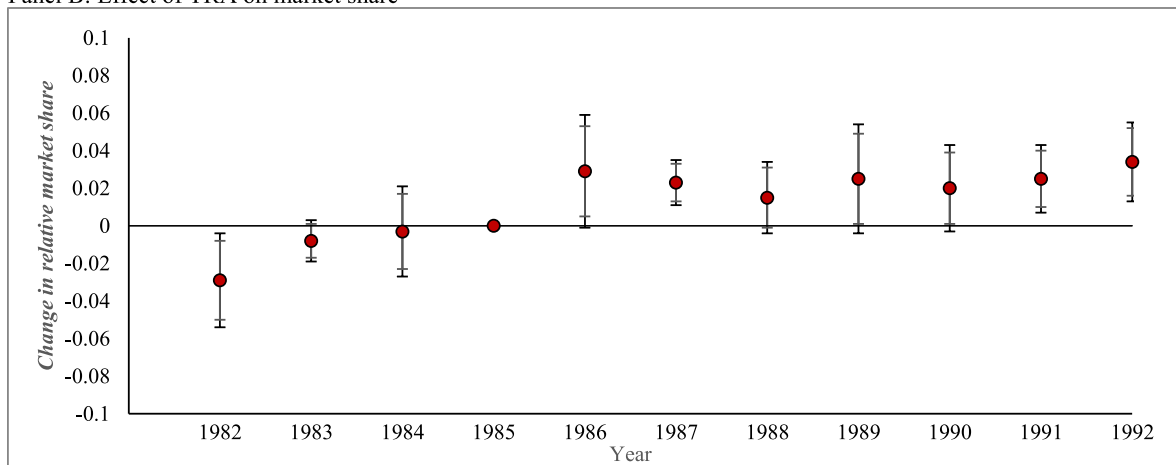
²³ Computed as $0.44\% = 0.042$ (from Table 4, column 2)/0.096 (decrease in MTR described above) and 0.34 percentage point $= 0.033$ (from Table 4, column 4)/0.096 (decrease in MTR described above).

²⁴ In additional analyses tabulated in the online appendix (in Table OA-14), we examine whether the TRA leads profitable airlines to lower prices in routes in which their rivals are financially constrained, regardless of whether they have a loss. We find that while profit-making airlines marginally lower prices post-TRA in routes where their financially constrained rivals have greater market share, the market share of profit-making airlines in these routes is unchanged post-TRA. A possible explanation for these results is that the TRA eases the financing constraints of airlines with positive tax burdens (e.g., Edwards et al., 2016; Dobridge, 2021), which enables them to fend off predatory threats from their profit-making rivals.

Panel A: Effect of TRA on ticket prices



Panel B: Effect of TRA on market share

**Fig. 1.** Dynamic effect of TRA on ticket prices and market share of profit-making relative to loss-making airline carriers

Panel A: Effect of TRA on ticket prices

Panel B: Effect of TRA on market share

Notes: In the figures above, the x-axis represents years centered around the enactment of the Tax Reform Act (TRA) of 1986. In Panel A, the y-axis represents the percent year-on-year change in the average ticket prices of profitable airlines compared to those of loss-making airlines in routes where loss-airlines have high market share, and in Panel B, the y-axis represents the change in the relative market share of profitable vs. loss airlines in each period. The Panel A figure corresponds to the regression tabulated in Column 2 in Table 4, Panel A. The Panel B figure corresponds to the regression tabulated in Column 4 in Table 4, Panel A. The figures also plot the two-tailed 90% and 95% confidence intervals around each point estimate.

5.2. Effect of TRA on entry and exit patterns

Having shown that the TRA leads profitable airlines to lower prices and gain market share from loss-making rivals, we next examine whether these competitive effects translate into changes along the extensive margin. If the tax cut strengthens the competitive position of profitable airlines relative to loss-making competitors, it should also affect firms' decisions to enter and exit markets. Specifically, after the TRA, profitable airlines should be more likely to enter routes where loss-making airlines are economically important, while loss-making airlines should be more likely to exit such routes. We test these predictions by estimating the following regression.

$$y_{r,t} = \beta_1 \text{MKT IMPORTANCE OF LOSS AIRLINES}_{r,t-1} + \beta_2 \text{MKT IMPORTANCE OF LOSS AIRLINES}_{r,t-1} \times \text{POST86}_t + \alpha_r + \alpha_t + \alpha_t \times \text{LN}(\text{DISTANCE}) + \mathbf{r}'\mathbf{X} + \varepsilon_{r,t} \quad (2)$$

Where r and t index routes and year-quarters, respectively; $y_{r,t}$ represents the number of airlines entering into or exiting from a route in a given period. We change our unit of analyses to the route-year-quarter level from the airline-route-year-quarter level because entry is not defined for incumbent airlines in a route and exit is not defined unless an airline is already operating in a route, making the airline-route level analyses unsuitable. Further, we no longer constrain our sample to routes with two or more airlines because the TRA

Table 5

Cross-sectional effects based on airline financing constraints.

Dependent Variable:	LN(PRICE)		MARKET SHARE	
	(1)	(2)	(3)	(4)
	Coef.	Coef.	Coef.	Coef.
	t-Stat.	t-Stat.	t-Stat.	t-Stat.
	Aggregate market share of financially constrained loss-making airlines			
	High	Low	High	Low
<i>MKT IMPORTANCE OF LOSS AIRLINES</i>	−0.013 (−1.080)	0.008 (0.684)	0.066*** (4.911)	0.041*** (5.627)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i>	0.025* (2.006)	−0.008 (−0.603)	−0.087*** (−6.001)	−0.074*** (−7.776)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>POST86</i>	0.039** (2.656)	0.009 (0.721)	−0.031** (−2.452)	−0.000 (−0.020)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>POST86</i>	−0.049*** (−3.385)	−0.019 (−1.247)	0.038** (2.712)	0.014 (1.240)
<i>p</i> Value for difference in coefficient for: <i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>PROFIT AIRLINE</i> × <i>POST86</i>	0.061		0.105	
Airline-route level time varying controls	Included	Included	Included	Included
Fixed effects				
Airline × Route	Included	Included	Included	Included
Airline × Year-Quarter	Included	Included	Included	Included
Year-Quarter × LN(DISTANCE)	Included	Included	Included	Included
R-Squared	90.2%	94.1%	86.6%	89.6%
No. of Observations	253,659	111,153	253,659	111,153

Notes: This table presents the results from regressions of 1) the natural logarithm of average ticket prices charged by airlines in each route-year-quarter or 2) airlines' market shares in each route-year-quarter on interaction terms between the aggregate market share of loss-making airlines operating in the route-year-quarter, an indicator variable that equals one for whether the airline made a tax profit the previous year (*PROFIT AIRLINE*), and an indicator variable that equals one for the post-1986 TRA period (*POST86*), and control variables. Note that the main effects of *POST86*, *PROFIT AIRLINE*, and *POST86* × *PROFIT AIRLINE* are subsumed by our fixed effects. We use weighted least squares when the dependent variable is ticket prices, with weights based on the average number of passengers traveling with an airline in a route over the entire sample period. The full sample is partitioned into two groups based on whether the aggregate market share of financially constrained loss-making airlines is in the top three quartiles or not. See [Appendix A](#) for variable definitions. Standard errors are clustered at the airline and year-quarter levels. ***, **, and * denote statistical significance at the two-tailed 1, 5, and 10 percent levels, respectively.

changes the threat of entry even in markets with only one (loss-making) incumbent. [Table 6](#), Panel A presents the descriptive statistics for the variables used in our analyses and Panel B presents the results from estimating equation (2).

In column 1 of Panel B, the dependent variable is *#ENTRY*, which measures the number of airlines entering a route. The coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* × *POST86* is 0.008 (t-stat. = 2.51), suggesting that markets where loss-making incumbents have relatively high market share see a significant increase in new entrants, following the TRA. Columns 2 and 3 break down *#ENTRY* into the number of profitable and loss-making airlines entering a market (*#ENTRY - PROFIT AIRLINES* and *#ENTRY - LOSS AIRLINES*, respectively; categories and variable names are defined in [Appendix A](#)). The coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* × *POST86* is 0.006 (t-stat. = 2.30) and 0.002 (t-stat. = 1.37) when the dependent variable is *#ENTRY - PROFIT AIRLINES* and *#ENTRY - LOSS AIRLINES*, respectively. The p-value for the difference between these coefficients is marginally insignificant at 0.138. These results tell us that the increase in entry into markets where loss-making incumbents have a significant presence is due to profitable airlines entering the market. In terms of economic magnitude, a one standard deviation increase in *MKT IMPORTANCE OF LOSS AIRLINES* leads to an 11% increase in the number of profitable airlines entering the market after the TRA.²⁵

Columns 4 to 6 in [Table 6](#), Panel B examine airline exit patterns. Column 4 shows that the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* × *POST86* is 0.015 (t-stat. = 2.14) when the dependent variable is *#EXITS*, suggesting that markets where loss-making incumbents are economically important see a significant increase in airline exits after the TRA. Further, the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* × *POST86* is −0.002 (t-stat. = −0.82) and 0.017 (t-stat. = 2.36) when the dependent variable is *#EXIT - PROFIT AIRLINES* and *#EXIT - LOSS AIRLINES*, respectively, and the p-value for the difference between these coefficients is 0.014. These results imply that the increase in exits is entirely driven by loss-making airlines. In economic terms, a one standard deviation increase in *MKT IMPORTANCE OF LOSS AIRLINES* leads to a 65% increase in the number of loss-making airlines exiting the market after the TRA.²⁶ The large economic magnitude of exits is partly a reflection of the low unconditional rate of exits (as discussed

²⁵ We compute the economic magnitude as 0.006 (the regression coefficient for *IMPORTANCE OF LOSS AIRLINES* × *POST86*) divided by 0.055 (the average value of *#ENTRY - PROFIT AIRLINES* in [Table 6](#), Panel A).

²⁶ We compute the economic magnitude as 0.017 (the regression coefficient for *IMPORTANCE OF LOSS AIRLINES* × *POST86*) divided by 0.026 (the average value of *#EXIT - LOSS AIRLINES*).

Table 6

Effect of tax rate reductions and tax loss status on airline entry into new routes and exit from old routes.

Panel A: Descriptive statistics						
Variables	Mean	SD	P25	P50	P75	N
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> ⁺	0.187	0.332	0.000	0.000	0.236	358,801
<i>#ENTRY</i>	0.073	0.276	0.000	0.000	0.000	358,801
<i>#ENTRY - PROFIT AIRLINES</i>	0.055	0.239	0.000	0.000	0.000	358,801
<i>#ENTRY - LOSS AIRLINES</i>	0.017	0.133	0.000	0.000	0.000	358,801
<i>#EXITS</i>	0.073	0.273	0.000	0.000	0.000	358,801
<i>#EXITS - PROFIT AIRLINES</i>	0.047	0.219	0.000	0.000	0.000	358,801
<i>#EXITS - LOSS AIRLINES</i>	0.026	0.162	0.000	0.000	0.000	358,801
<i>%CONNECTING - ROUTE</i>	0.806	0.357	0.848	1.000	1.000	358,801
<i>LN(POPULATION)</i>	14.229	0.891	13.604	14.153	14.687	358,801
<i>LN(INCOME)</i>	9.740	0.192	9.600	9.755	9.886	358,801
Panel B: Regression results						
Dependent Variable:	<i>#ENTRY</i>	<i>#ENTRY - PROFIT AIRLINES</i>	<i>#ENTRY - LOSS AIRLINES</i>	<i>#EXITS</i>	<i>#EXITS - PROFIT AIRLINES</i>	<i>#EXITS - LOSS AIRLINES</i>
	(1)	(2)	(3)	(4)	(5)	(6)
	Coef.	Coef.	Coef.	Coef.	Coef.	Coef.
	t-Stat.	t-Stat.	t-Stat.	t-Stat.	t-Stat.	t-Stat.
<i>MKT IMPORTANCE OF LOSS AIRLINES</i>	−0.002 (−0.652)	0.006** (2.682)	−0.008*** (−4.457)	0.002 (0.547)	−0.008*** (−6.067)	0.010*** (3.379)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>POST86</i>	0.008** (2.514)	0.006** (2.296)	0.002 (1.372)	0.015** (2.135)	−0.002 (−0.818)	0.017** (2.359)
<i>%CONNECTING - ROUTE</i>	0.012* (1.755)	0.013** (2.036)	−0.001 (−0.303)	−0.006 (−0.979)	−0.003 (−0.654)	−0.003 (−0.986)
<i>LN(POPULATION)</i>	0.042 (1.178)	0.044** (2.170)	−0.002 (−0.082)	−0.042 (−1.515)	−0.077** (−2.425)	0.035 (1.602)
<i>LN(INCOME)</i>	0.223*** (3.711)	0.145*** (3.098)	0.078** (2.409)	0.240*** (3.169)	0.081** (2.514)	0.159** (2.141)
<i>p</i> Value for difference in coefficient for: <i>MKT IMPORTANCE OF LOSS AIRLINES</i> × <i>POST86</i>			0.138			0.014
<i>Fixed effects</i>						
Route	Included	Included	Included	Included	Included	Included
Year-Quarter	Included	Included	Included	Included	Included	Included
Year-Quarter × <i>LN(DISTANCE)</i>	Included	Included	Included	Included	Included	Included
R-Squared	16.0%	16.2%	11.7%	16.0%	16.2%	11.7%
No. of Observations	358,801	358,801	358,801	358,801	358,801	358,801

Notes: Panel A presents the descriptive statistics for the variables used in the entry/exit analyses. In Panel A, + indicates that the variable is standardized to have a mean of zero and a standard deviation of one in our regression analyses. Panel B presents the results from regressions of the number of profitable and loss-making airlines that enter and exit a route each year-quarter on the aggregate market share of loss-making airlines operating in the route-year-quarter, its interaction with an indicator variable that equals one for the post-1986 TRA period (*POST86*), and control variables. The main effect of *POST86* is subsumed by our fixed effects. See Appendix A for variable definitions. Standard errors are clustered at the route and year-quarter levels. ***, **, and * denote statistical significance at the two-tailed 1, 5, and 10 percent levels, respectively. Note that the sample size in this table is different than that in Tables 4 and 5 because 1) the unit of observation is at the route-year-quarter level (rather than airline-route-year-quarter level), and 2) we do not constrain the sample to routes that have at least two airlines.

in section 2.3). In other words, for every 10-percentage point increase in *MKT IMPORTANCE OF LOSS AIRLINES*, profitable firms' entry increases by 3.3%, while loss-making firms' exit increases by 19.7%. Collectively, the evidence in Table 6 suggests that the TRA leads to a significant increase in the number of profitable airlines entering markets where loss-making incumbents have high market share, and a simultaneous increase in the number of loss-making airlines exiting those very same markets.²⁷

To explore further, we examine whether profitable airlines are more likely to enter markets in which the incumbent airlines are not only loss-making but also financially constrained. We re-estimate equation (2) after partitioning our sample into route-year-quarters in which financially constrained loss-making incumbents have high vs. low market share. Consistent with our prediction, columns 1 and 2 in Table 7, Panel A show that, after the TRA, airlines are almost four times as likely to enter markets in which the financially

²⁷ If an airline experiences a decline in ticket sales, an alternative to exiting the route altogether might be for the airline to shift to a smaller plane that is more fuel efficient. While we lack data on the aircraft types used by airlines during our sample period, we consulted an expert on the airline industry who indicated that such a switch was unlikely because airline management is incentivized to optimize a metric known as Cost Per Available Seat Mile (CASM), which would be adversely impacted if airlines responded to shrinking demand by switching to smaller planes.

Table 7

Effect of tax rate reductions and tax loss status on airline entry into markets conditioning on financial constraints.

Panel A: Cross-sectional variation in financing constraints of incumbent loss-making airlines						
Dependent Variable:	#ENTRY		#ENTRY - PROFIT AIRLINES		#ENTRY - LOSS AIRLINES	
	(1)	(2)	(3)	(4)	(5)	(6)
	Coef.	Coef.	Coef.	Coef.	Coef.	Coef.
	t-Stat.	t-Stat.	t-Stat.	t-Stat.	t-Stat.	t-Stat.
	Aggregate market share of financially constrained loss-making airlines					
	HIGH	LOW	HIGH	LOW	HIGH	LOW
MKT IMPORTANCE OF LOSS AIRLINES	0.002 (0.172)	−0.003 (−1.070)	0.013 (1.603)	0.003 (1.399)	−0.011** (−2.193)	−0.006*** (−3.242)
MKT IMPORTANCE OF LOSS AIRLINES × POST86	0.039*** (3.852)	0.010** (2.635)	0.030*** (3.348)	0.008*** (3.109)	0.009** (2.149)	0.002 (0.823)
%CONNECTING - ROUTE	0.026** (2.123)	0.010 (1.272)	0.019 (1.674)	0.011* (1.776)	0.007 (1.244)	−0.002 (−0.451)
LN(POPULATION)	0.185** (2.475)	0.028 (0.792)	0.163*** (3.238)	0.031 (1.415)	0.022 (0.397)	−0.004 (−0.163)
LN(INCOME)	0.601*** (3.998)	0.188*** (3.766)	0.429*** (3.188)	0.105** (2.585)	0.172** (2.471)	0.083*** (2.750)
p Value for difference in coefficient for: MKT IMPORTANCE OF LOSS AIRLINES × POST86	0.005		0.025		0.030	
Fixed effects						
Route	Included	Included	Included	Included	Included	Included
Year-Quarter	Included	Included	Included	Included	Included	Included
Year-Quarter × LN(DISTANCE)	Included	Included	Included	Included	Included	Included
R-Squared	11.4%	8.1%	10.6%	6.5%	8.9%	7.2%
No. of Observations	77,597	281,204	77,597	281,204	77,597	281,204
Panel B: Cross-sectional variation in financing constraints of airline entering new market						
Dependent Variable:	#ENTRY - PROFIT FC AIRLINES	#ENTRY - PROFIT NOT FC AIRLINES	#ENTRY - LOSS FC AIRLINES	#ENTRY - LOSS NOT FC AIRLINES		
	(1)	(2)	(3)	(4)		
	Coef.	Coef.	Coef.	Coef.		
	t-Stat.	t-Stat.	t-Stat.	t-Stat.		
MKT IMPORTANCE OF LOSS AIRLINES	−0.001* (−1.845)	0.007*** (3.883)	−0.004** (−2.501)	−0.003*** (−6.217)		
MKT IMPORTANCE OF LOSS AIRLINES × POST86	0.006*** (4.012)	0.001 (0.228)	0.001 (0.812)	0.001 (1.056)		
%CONNECTING - ROUTE	0.008 (1.466)	0.005 (1.296)	0.001 (0.378)	−0.001 (−0.896)		
LN(POPULATION)	0.023* (1.986)	0.021 (1.430)	0.026 (1.177)	−0.028** (−2.217)		
LN(INCOME)	0.068** (2.510)	0.077** (2.271)	0.018 (0.850)	0.059** (2.459)		
p Value for difference in coefficient for: MKT IMPORTANCE OF LOSS AIRLINES × POST86	0.069		0.778			
Fixed effects						
Route	Included	Included	Included	Included		
Year-Quarter	Included	Included	Included	Included		
Year-Quarter × LN(DISTANCE)	Included	Included	Included	Included		
R-Squared	5.4%	4.7%	5.3%	4.5%		
No. of Observations	358,801	358,801	358,801	358,801		

Notes: Panel A in this table presents the results from regressions of the number of profitable and loss-making airlines that enter a route each year-quarter on interaction terms between the aggregate market share of loss-making airlines operating in the route-year-quarter, its interaction with an indicator variable that equals one for the post-1986 TRA period (*POST86*), and control variables. In Panel A, the full sample is partitioned into two groups based on whether the aggregate market share of financially constrained loss-making airlines is in the top three quartiles or not. Panel B presents results from regressions of the number of airlines that enter a route each year-quarter conditioning on the airline's tax status and financial constraints, on the aggregate market share of loss-making airlines operating in the route-year-quarter, its interaction with an indicator variable that equals one for the post-1986 TRA period (*POST86*), and control variables. The main effect of *POST86* is subsumed by our fixed effects. See [Appendix A](#) for variable definitions. Standard errors are clustered at the route and year-quarter levels. ***, **, and * denote statistical significance at the two-tailed 1, 5, and 10 percent levels, respectively. Note that the sample size in this table is different than that in [Tables 4 and 5](#) because 1) the unit of observation is at the route-year-quarter level (rather than airline-route-year-quarter level), and 2) we do not constrain the sample to routes that have at least one profitable and at least one loss-making airline.

constrained loss-making incumbents have high market share. Specifically, when the dependent variable is *#ENTRY*, the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* $\times$ *POST86* is 0.039 (t-stat. = 3.85) in routes in which the financially constrained loss-making incumbents have high market share compared to a coefficient of 0.010 (t-stat. = 2.64) in routes where the financially constrained loss-making incumbents have low market share. Columns 3 to 6 decompose *#ENTRY* into *#ENTRY - PROFIT AIRLINES* and *#ENTRY - LOSS AIRLINES*. Consistent with the evidence in Table 6, post-TRA, profitable airlines are more likely to enter routes where loss-making incumbents have high market share, on average. Further, the number of profitable airlines entering a market post-TRA is 3.75 times greater when the financially constrained loss-making incumbents have high market share.

The analyses above exploit cross-sectional variation in the characteristics of *incumbent* airlines (loss vs. profit and financially constrained vs. unconstrained). Next, we exploit cross-sectional variation in these same characteristics of *new entrants*. To the extent the cash tax savings accompanying the TRA ease financing constraints for profitable airlines, we posit that such financially constrained profitable airlines are more likely to respond to the TRA by preying on their loss-making rivals. In contrast, we predict that financially unconstrained profitable airlines are less likely to prey on their rivals in response to the TRA because they likely could have engaged in such behavior prior to the TRA (the cash shock from the TRA does not materially change their behavior in this regard).

Consistent with our prediction, Table 7, Panel B shows that the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* $\times$ *POST86* is positive and significant only in column 1, where the dependent variable is the number of financially constrained profitable airlines entering a market (*#ENTRY - PROFIT FC AIRLINES*). Also consistent with our prediction, columns 2 to 4 show that the coefficient for *MKT IMPORTANCE OF LOSS AIRLINES* $\times$ *POST86* is insignificant when the dependent variable is the number of 1) financially unconstrained profitable airlines entering a market, 2) financially constrained loss airlines entering a market, and 3) financially unconstrained loss airlines entering a market.

Finally, we examine in greater detail the characteristics of airlines that exit markets following the TRA. As discussed earlier, financially constrained firms are more vulnerable to predation because they typically lack the liquidity needed to withstand cash flow reductions arising from a price war. Accordingly, we predict that the TRA leads to an increase in exits among financially constrained, loss-making airlines. Table 8 presents results from six regressions. We first decompose market exits into those by financially constrained airlines and those by unconstrained airlines. We then further decompose exits into 1) profitable airlines that are financially constrained versus unconstrained and 2) loss-making airlines that are financially constrained versus unconstrained. We find that the TRA increases exits only among airlines that are both loss-making and financially constrained (column 5).

Table 8

Effect of tax rate reductions on airline exits by financial constrained vs. unconstrained incumbent airlines.

Dependent Variable:	#EXITS - FC AIRLINES	#EXITS - NOT FC AIRLINES	#EXITS - PROFIT FC AIRLINES	#EXITS - PROFIT NOT FC AIRLINES	#EXITS - LOSS FC AIRLINES	#EXITS - LOSS NOT FC AIRLINES
	(1)	(2)	(3)	(4)	(5)	(6)
	Coef.	Coef.	Coef.	Coef.	Coef.	Coef.
	t-Stat.	t-Stat.	t-Stat.	t-Stat.	t-Stat.	t-Stat.
<i>MKT IMPORTANCE OF LOSS AIRLINES</i>	-0.001 (-0.302)	0.003 (1.442)	-0.003*** (-5.829)	-0.006*** (-4.226)	0.002 (0.572)	0.009*** (6.693)
<i>MKT IMPORTANCE OF LOSS AIRLINES</i> $\times$ <i>POST86</i>	0.012* (1.714)	0.002 (0.853)	-0.002 (-1.533)	0.000 (0.020)	0.015** (2.095)	0.002 (0.926)
%CONNECTING - ROUTE	-0.001 (-0.513)	-0.005 (-1.024)	-0.001 (-0.596)	-0.002 (-0.532)	-0.000 (-0.096)	-0.003 (-1.389)
LN(POPULATION)	0.006 (0.301)	-0.048*** (-3.058)	-0.020 (-0.966)	-0.056*** (-3.864)	0.027** (2.558)	0.008 (0.487)
LN(INCOME)	0.144* (2.018)	0.096** (2.669)	0.050*** (3.042)	0.031 (0.991)	0.094 (1.303)	0.065*** (3.551)
<i>p</i> Value for difference in coefficient for: <i>MKT IMPORTANCE OF LOSS AIRLINES</i> $\times$ <i>POST86</i>	0.194		0.336		0.097	
Fixed effects						
Route	Included	Included	Included	Included	Included	Included
Year-Quarter	Included	Included	Included	Included	Included	Included
Year-Quarter $\times$ LN(DISTANCE)	Included	Included	Included	Included	Included	Included
R-Squared	14.5%	13.3%	17.4%	13.2%	11.2%	11.4%
No. of Observations	358,801	358,801	358,801	358,801	358,801	358,801

Notes: This table presents the results from regressions of the number of airlines that exit a route each year-quarter on the aggregate market share of loss-making airlines operating in the route-year-quarter, its interaction with an indicator variable that equals one for the post-1986 TRA period (*POST86*), and control variables. The main effect of *POST86* is subsumed by our fixed effects. We partition exits by whether an airline is profitable or loss-making, and whether it is financially constrained or not. See Appendix A for variable definitions. Standard errors are clustered at the route and year-quarter levels. ***, **, and * denote statistical significance at the two-tailed 1, 5, and 10 percent levels, respectively. Note that the sample size in this table is different than that in Tables 4 and 5 because 1) the unit of observation is at the route-year-quarter level (rather than airline-route-year-quarter level), and 2) we do not constrain the sample to routes that have at least one profitable and at least one loss-making airline.

Collectively, our analyses of entry and exit patterns of airlines following the passage of the TRA show that the TRA and associated cash tax effects for profitable firms help financially constrained profitable airlines gain a competitive advantage over their financially constrained loss-making rivals.

5.3. Price reversals after the exit of loss-making airlines

In this section, we examine whether profitable airlines change their pricing policies after the exit of their loss-making rivals from a route. Theory suggests predators will raise prices after their rivals exit the market. Specifically, predators initially lower prices to reduce industry cash flows and force the exit of their rivals, and subsequently increase prices to earn monopoly rents after rivals exit the market. To the extent profitable airlines did indeed lower prices after the TRA to prey on their loss-making rivals, the exits of these rivals should encourage profitable airlines to raise prices. We test the above prediction by estimating the following regression.

$$y_{i,r,t} = \beta_1 \#Proxy\ for\ exits_{r,t-1} + \beta_2 \#Proxy\ for\ exits_{r,t-1} \times PROFIT\ AIRLINE_{i,t-1} + \beta_3 \#Proxy\ for\ exits_{r,t-1} \times POST89_t + \beta_4 \#Proxy\ for\ exits_{r,t-1} \times PROFIT\ AIRLINE_{i,t-1} \times POST89_t + \alpha_i \times \alpha_r + \alpha_i \times \alpha_t + \alpha_t \times LN(DISTANCE) + \tau'X + \varepsilon_{i,r,t} \quad (3)$$

Where i , r , and t index airlines, routes, and year-quarters, respectively; $y_{i,r,t}$ is the natural logarithm of the average fare charged by an airline in a route-year-quarter ($LN(PRICE)$). $\#Proxy\ for\ exits$ is one of three variables: 1) $\#PAST\ EXITS$, the total number of airlines that exit a route in the previous 12 quarters, 2) $\#PAST\ EXITS\ BY\ PROFIT\ AIRLINES$, the number of profitable airlines that exit a route in the previous 12 quarters, and 3) $\#PAST\ EXITS\ BY\ LOSS\ AIRLINES$, the number of loss-making airlines that exit a route in the previous 12 quarters.²⁸ $POST89$ is an indicator variable that equals one for calendar years 1989 and onwards (i.e., 12 quarters after the TRA passed the House of Representatives in December 1985). The other variables are as defined earlier. We estimate equation (3) using weighted least squares (as we did for equation (1)). Table 9 presents the results.²⁹

In column 1, we examine the effect of post-TRA airline exits ($\#PAST\ EXITS$) on the pricing policies of profitable airlines, regardless of the tax status of the exiting airlines. We find that the coefficient for $\#PAST\ EXITS \times PROFIT\ AIRLINE \times POST89$ is insignificant, indicating that profitable airlines do not differentially alter their pricing policy in response to the exit of the average airline. Columns 2 and 3 decompose $\#PAST\ EXITS$ into $\#PAST\ EXITS\ BY\ PROFIT\ AIRLINES$ and $\#PAST\ EXITS\ BY\ LOSS\ AIRLINES$ based on the tax status of exiting airlines. Column 2 shows that the coefficient for $\#PAST\ EXITS\ BY\ PROFIT\ AIRLINES \times PROFIT\ AIRLINE \times POST89$ is insignificant, and in contrast, column 3 shows that the coefficient for $\#PAST\ EXITS\ BY\ LOSS\ AIRLINES \times PROFIT\ AIRLINE \times POST89$ is significantly positive (coef. = 0.014; t-stat. = 2.39). The results in columns 2 and 3 suggest profitable airlines raise prices after the exit of their loss-making competitors in the post-TRA period but do not change their response to the exit of their profitable competitors. Column 4 in the table shows that our results are robust to simultaneously including both $\#PAST\ EXITS\ BY\ PROFIT\ AIRLINES$ and $\#PAST\ EXITS\ BY\ LOSS\ AIRLINES$ in our regression. Economically, a one standard deviation increase in $\#PAST\ EXITS\ BY\ LOSS\ AIRLINES$ leads to a 1.4% increase in the ticket prices of profitable airlines after the TRA. Put differently, each additional exit of a loss airline is associated with an additional 2.5% increase in prices charged by profitable firms, which corresponds to \$4.10 price increase.³⁰ These findings are consistent with our hypothesis that profitable airlines engage in a predation strategy post-TRA to accelerate the exit of loss-making airlines and collect rents by raising prices after their exit.³¹

6. Conclusion

This paper examines whether tax changes have disparate product market consequences for firms with high versus low tax burdens. Tax cuts typically benefit firms with higher tax burdens by increasing their after-tax cash flows while providing little relief to firms with low current tax burdens. As a result, tax cuts can provide a competitive advantage to firms that pay high taxes relative to those that pay low taxes. We predict that profitable firms use the cash savings from the tax cut to lower prices and gain market share from their loss-making rivals, especially when their loss-making rivals are financially constrained and lack sufficient liquidity to survive a prolonged price war.

To test our prediction, we use detailed price and quantity data from the airline industry around the passage of the Tax Reform Act (TRA) in 1986, which cut the highest statutory tax rate by 12 percentage points. Holding the route and period constant, we find that the TRA is followed by a 4.2% relative decrease in the ticket prices of profitable airlines compared to those of loss-making airlines, and a 3.3 percentage point increase in the market share of profitable airlines. These results manifest in routes where loss-making airlines are economically important and have high aggregate market share. Further, the effect of the TRA on the relative ticket prices and market

²⁸ We assume that it takes loss-making airlines three years on average to exit a route in response to a rival airline's predatory pricing strategy. Our results are robust to assuming a 2-year period for such exits to occur.

²⁹ Note that our sample size changes again for two reasons: 1) the sample period for this analysis begins in 1985 (rather than 1982) since we require data on the number of exits in the preceding 12 quarters, and 2) we do not restrict our sample to route-year-quarters with at least two airlines (because our interest lies in examining price responses to the exit of loss-making airlines).

³⁰ The average and standard deviation of $\#PAST\ EXITS\ BY\ LOSS\ AIRLINES$ are 0.30 and 0.565, respectively.

³¹ To corroborate our inferences, we also use an entirely different setting to test our research question. In our online appendix, we examine how changes in state corporate income tax rates affect the market share of profit and loss-firms operating in the state. See our online appendix for a more complete discussion of the benefits and limitations of this approach and the results of the state-level tests (which are consistent with our results in the airline industry).

Table 9

Price reversals after the exit of loss-making airlines.

Dependent Variable:	LN(PRICE)			
	(1)	(2)	(3)	(4)
	Coef.	Coef.	Coef.	Coef.
	t-Stat.	t-Stat.	t-Stat.	t-Stat.
#PAST EXITS	−0.009* (−1.874)			
#PAST EXITS × POST89	0.006 (0.844)			
#PAST EXITS × PROFIT AIRLINE	−0.002 (−0.622)			
#PAST EXITS × PROFIT AIRLINE × POST89	0.002 (0.250)			
#PAST EXITS BY PROFIT AIRLINES		−0.004 (−1.103)		−0.005 (−1.250)
#PAST EXITS BY PROFIT AIRLINES × POST89		0.004 (0.439)		0.004 (0.414)
#PAST EXITS BY PROFIT AIRLINES × PROFIT AIRLINE		0.003 (0.893)		0.003 (0.952)
#PAST EXITS BY PROFIT AIRLINES × PROFIT AIRLINE × POST89		−0.005 (−0.629)		−0.005 (−0.651)
#PAST EXITS BY LOSS AIRLINES			−0.008 (−1.620)	−0.008 (−1.589)
#PAST EXITS BY LOSS AIRLINES × POST89			0.006 (0.974)	0.006 (0.907)
#PAST EXITS BY LOSS AIRLINES × PROFIT AIRLINE			−0.013** (−2.274)	−0.013** (−2.278)
#PAST EXITS BY LOSS AIRLINES × PROFIT AIRLINE × POST89			0.014** (2.392)	0.014** (2.332)
#AIRLINES	−0.014*** (−3.619)	−0.013*** (−3.296)	−0.013*** (−3.493)	−0.014*** (−3.672)
%CONNECTING - AIRLINE	−0.008 (−0.332)	−0.008 (−0.327)	−0.008 (−0.320)	−0.008 (−0.314)
%CONNECTING - ROUTE	0.092*** (6.195)	0.092*** (6.265)	0.091*** (6.214)	0.091*** (6.195)
LN(POPULATION)	−0.518** (−2.658)	−0.514** (−2.645)	−0.531** (−2.760)	−0.528** (−2.716)
LN(INCOME)	0.223 (0.846)	0.219 (0.835)	0.227 (0.876)	0.233 (0.902)
<i>Fixed effects</i>				
Airline × Route	Included	Included	Included	Included
Airline × Year-Quarter	Included	Included	Included	Included
Year-Quarter × LN(DISTANCE)	Included	Included	Included	Included
R-Squared	90.4%	90.4%	90.4%	90.4%
No. of Observations	634,861	634,861	634,861	634,861

Notes: This table presents the results from regressions of the natural logarithm of average ticket prices charged by airlines in each route-year-quarter on the number of airlines that exit a route in the previous 12 quarters, an indicator variable that equals one for whether the airline made a tax profit the previous year, an indicator variable that equals one for calendar years 1989 and onwards, interaction terms between these variables and control variables. We use weighted least squares based on the average number of passengers traveling with an airline in a route over the entire sample period. See [Appendix A](#) for variable definitions. Standard errors are clustered at the airline and year-quarter levels. ***, **, and * denote statistical significance at the two-tailed 1, 5, and 10 percent levels, respectively.

share of profitable vs. loss airlines is greater when loss-making rivals are financially constrained.

On the extensive margin, the TRA leads to an 11% increase in the number of profitable airlines entering markets where loss-making incumbents have high aggregate market share, and a 65% increase in the number of loss-making airlines exiting these very same markets. Further, profitable airlines enter in greater numbers into markets where the loss-making incumbents are financially constrained. Finally, we find that profitable airlines raise ticket prices following the exit of their loss-making competitors in the post-TRA period.

We caveat that while airline data are rich and enable strong empirical tests, the inferences we draw from studying the airline industry might not generalize to the broader economy. In addition, while we interpret our results to be consistent with the predation hypothesis of [Bolton and Scharfstein \(1990\)](#), our inability to observe marginal costs at the airline-route level makes it difficult to distinguish a predation strategy from one where a profitable airline uses its cash tax savings to grow market share, and such growth comes from markets in which their rivals are incurring losses. Notwithstanding these caveats, our study contributes both to our understanding of the heterogeneous consequences of tax law changes and to our understanding of product market behavior in response to taxation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variable Definitions

This table provides a detailed description of the procedure used to compute each variable used in our analyses. All continuous variables except for the airline entry/exit variables (e.g., *#ENTRY*) are winsorized at 1% and 99% of their distribution. The variables are listed in alphabetical order.

Variable	Definition
<i>#AIRLINES</i>	The number of airlines in route <i>r</i> in year-quarter <i>t</i> . The data is obtained from the DB1A database (http://data.nber.org/data/dot-db1a/).
<i>#ENTRY</i>	The number of new entries in route <i>r</i> in year-quarter <i>t</i> . If an airline has not operated in the route for the previous 8 quarters and starts to operate in year-quarter <i>t</i> , it is regarded as an entry. Entries triggered by airline mergers are excluded.
<i>#ENTRY - LOSS AIRLINES</i>	The number of new entries of tax-loss airlines (i.e., <i>PROFIT AIRLINE</i> = 0) in route <i>r</i> in year-quarter <i>t</i> .
<i>#ENTRY - PROFIT AIRLINES</i>	The number of new entries of tax-profit airlines (i.e., <i>PROFIT AIRLINE</i> = 1) in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS</i>	The number of exits of airlines in route <i>r</i> in year-quarter <i>t</i> . If an airline exits the route in year-quarter <i>t</i> and doesn't operate in the next 8 quarters, it is regarded as an exit. Exits triggered by airline mergers or cessation are excluded.
<i>#EXITS - LOSS AIRLINES</i>	The number of exits of tax-loss airlines (i.e., <i>PROFIT AIRLINE</i> = 0) in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - LOSS FC AIRLINES</i>	The number of exits of airlines that were tax-loss and were financially constrained (i.e., <i>FIN CONSTRAINED</i> = 1) in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - LOSS NOT FC AIRLINES</i>	The number of exits of airlines that were tax-loss and were not financially constrained (i.e., <i>FIN CONSTRAINED</i> = 0) in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - PROFIT AIRLINES</i>	The number of exits of tax-profit airlines (i.e., <i>PROFIT AIRLINE</i> = 1) in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - PROFIT FC AIRLINES</i>	The number of exits of airlines that were tax-profit and were financially constrained in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - PROFIT NOT FC AIRLINES</i>	The number of exits of airlines that were tax-profit and were not financially constrained in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - FC AIRLINES</i>	The number of exits of airlines that were financially constrained in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> .
<i>#EXITS - NOT FC AIRLINES</i>	The number of exits of airlines that were not financially constrained in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> .
<i>%CONNECTING - AIRLINE</i>	The share of passengers traveling connect for airline <i>j</i> in route <i>r</i> in year-quarter <i>t</i> (airline-route-year-quarter level). The data is obtained from the DB1A database.
<i>%CONNECTING - ROUTE</i>	The share of passengers traveling connect in route <i>r</i> in year-quarter <i>t</i> (route-year-quarter level). The data is obtained from the DB1A database.
<i>AVG. PASSENGERS</i>	The average number of passengers for airline <i>j</i> in route <i>r</i> over the sample period (1982 – 1992). The data is obtained from the DB1A database.
<i>FIN CONSTRAINED (airline)</i>	An indicator that equals one if an airline did not pay dividends (data DVC) in any of the previous three fiscal years and its lag cash (data CH) scaled by lag total assets (data AT) was in the low two terciles. Ranks are defined by year-quarter.
<i>LN(DISTANCE)</i>	The natural log (distance+1) of the route <i>r</i> . The data is obtained from the DB1A database.
<i>LN(INCOME)</i>	The natural log of the arithmetic average of income per-capita across the core-based statistical areas (CBSAs) at the two endpoints of route <i>r</i> in year <i>t</i> . The data is obtained from the Bureau of Economic Analysis (https://apps.bea.gov/regional/downloadzip.cfm). To identify the FIPS code for each airport, Azar et al. (2018)'s Airport-CBSA Xwalk data and Severin Borenstein's airport list data are used. If the income data is available only for one of the endpoints, the natural log of that income per-capita is used.
<i>LN(POPULATION)</i>	The natural log of the arithmetic average of the population across the core-based statistical areas at the two endpoints of route <i>r</i> in year <i>t</i> . The data is obtained from the Bureau of Economic Analysis (https://apps.bea.gov/regional/downloadzip.cfm). To identify the FIPS code for each airport, Azar et al. (2018)'s Airport-CBSA Xwalk data and Severin Borenstein's airport list data are used. If the population data is available only for one of the endpoints, the natural log of that population is used.
<i>LN(PRICE)</i>	The natural log of the average ticket prices of airline <i>j</i> in route <i>r</i> in year-quarter <i>t</i> . The data is obtained from the DB1A database.
<i>MARKET SHARE</i>	The market share of airline <i>j</i> in route <i>r</i> in year-quarter <i>t</i> . Market share is defined as the number of passengers of airline <i>j</i> in route <i>r</i> in year-quarter <i>t</i> scaled by the number of passengers in route <i>r</i> in year-quarter <i>t</i> . The data is obtained from the DB1A database.
<i>MKT IMPORTANCE OF LOSS AIRLINES</i>	The previous quarter's aggregated market share of airlines that were in tax-loss positions in the previous fiscal year in route <i>r</i> in year-quarter <i>t</i> . Market share is defined as the number of passengers of airline <i>j</i> in route <i>r</i> in year-quarter <i>t</i> scaled by the number of passengers in route <i>r</i> in year-quarter <i>t</i> . The data is obtained from the DB1A database.
<i>#PAST EXITS</i>	The number of exits of airlines in route <i>r</i> in the previous 12 quarters.
<i>#PAST EXITS BY LOSS AIRLINES</i>	The number of exits of tax-loss airlines (i.e., <i>PROFIT AIRLINE</i> = 0) in route <i>r</i> in the previous 12 quarters.
<i>#PAST EXITS BY PROFIT AIRLINES</i>	The number of exits of tax-profit airlines (i.e., <i>PROFIT AIRLINE</i> = 1) in route <i>r</i> in the previous 12 quarters.
<i>POST86</i>	An indicator that equals one for the calendar year 1986 and beyond.
<i>POST89</i>	An indicator that equals one for the calendar year 1989 and beyond.
<i>PROFIT AIRLINE</i>	An indicator that equals one if an airline was <i>not</i> in a tax loss position in the previous fiscal year. An airline is not in a tax-loss position in a given year if it has non-negative pretax income (data PI) and/or zero net loss carryforward (data TLCF).
<i>YEAR82</i>	An indicator that equals one for the calendar year 1982.
<i>YEAR83</i>	An indicator that equals one for the calendar year 1983.
<i>YEAR84</i>	An indicator that equals one for the calendar year 1984.

(continued on next page)

(continued)

Variable	Definition
YEAR86	An indicator that equals one for the calendar year 1986.
YEAR87	An indicator that equals one for the calendar year 1987.
YEAR88	An indicator that equals one for the calendar year 1988.
YEAR89	An indicator that equals one for the calendar year 1989.
YEAR90	An indicator that equals one for the calendar year 1990.
YEAR91	An indicator that equals one for the calendar year 1991.
YEAR92	An indicator that equals one for the calendar year 1992.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jacceco.2026.101873>.

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